

# NEBULA REPORT

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# 1. Executive Summary

## 1.1 Project scope and demonstrated capability

Nebula is a Stage-1 automated, simulation-in-the-loop analog receiver design-space exploration framework. It integrates parameterized CTLE design, real-SPICE evaluation, S-parameter channel modelling, behavioral one-tap DFE analysis, PPO and baseline optimizers, staged metric evaluation, caching, provenance, checkpoints, and reporting. The original engineering problem is to explore interacting receiver parameters without treating a nominal simulation or an optimizer score as proof of a complete physical receiver.

The implemented circuit is a one-stage, source-degenerated SKY130 NMOS CTLE. Its output is evaluated through a behavioral one-tap DFE so that receiver-level eye metrics can be measured after a four-port channel model. The project combines this electrical model with an AutoCkt-style PPO search agent, Random Search, and the Cross-Entropy Method (CEM).

[VERIFIED FROM SOURCE CODE] The primary implementation is distributed across `circuits/`, `simulator/`, `rl/`, `experiments/`, `analysis/`, and `tests/`. The principal orchestration entry point is `experiments/run_autockt_pipeline.py`; the core evaluation function is `simulator.receiver.evaluate_receiver()`.

The system's design objective is not simply to produce a high reward. A candidate must survive a sequence of electrical and receiver-level checks, and the final report distinguishes PASS, FAIL, and NOT CLAIMED. This last status is important: an unmeasured metric is not converted into a passing metric by a default value.

## 1.2 Evidence summary

| Metric                   | Value                 | Unit            | Source                         | Verification Status                 | Interpretation                                                            |
|--------------------------|-----------------------|-----------------|--------------------------------|-------------------------------------|---------------------------------------------------------------------------|
| Current regression suite | 632 passed, 2 skipped | tests           | Live pytest run                | [VERIFIED FROM LOGS]                | All executed repository tests passed; two were skipped.                   |
| v3 training updates      | 100                   | updates         | <code>results/v3_tt_1c</code>  | [VERIFIED FROM RAW SIMULATION DATA] | A completed real-backend training artifact exists.                        |
| v3 training evaluations  | 692                   | evaluations     | Same summary                   | [VERIFIED FROM RAW SIMULATION DATA] | Real-SPICE evaluation count recorded by the artifact.                     |
| Historical qualification | 36/36                 | conditions      | <code>docs/V3_QUALIFIC</code>  | [VERIFIED FROM LOGS]                | Candidate-fidelity TT/SS/FF qualification for pipeline_ep0.               |
| Fair PPO comparison      | 0/20                  | successes       | <code>results/benchmark</code> | [VERIFIED FROM LOGS]                | One-seed, budget-matched result; not evidence of general PPO superiority. |
| Whole-circuit area       | Not available         | mm <sup>2</sup> | <code>analysis/final_s</code>  | [NOT AVAILABLE]                     | Only partial MOS channel area is computed.                                |

The following nominal values are reported because they are present in the fresh real-SPICE evidence package, not because they are representative of every candidate or every operating condition:

| Metric               | Value  | Unit  | Source           | Verification Status                 | Interpretation                                                    |
|----------------------|--------|-------|------------------|-------------------------------------|-------------------------------------------------------------------|
| Eye height           | 1.55   | V     | results/llm_wrap | [VERIFIED FROM RAW SIMULATION DATA] | Locked-phase receiver eye result for that run.                    |
| Eye width            | 0.73   | UI    | Same             | [VERIFIED FROM RAW SIMULATION DATA] | Unit-interval width for the measured eye.                         |
| HD3                  | -75.62 | dB    | Same             | [VERIFIED FROM RAW SIMULATION DATA] | Measured at the documented HD3 test condition.                    |
| Input-referred noise | 0.372  | mVrms | Same             | [VERIFIED FROM RAW SIMULATION DATA] | Integrated noise over the implemented noise-analysis band.        |
| Observed DFE errors  | 0      | count | Same             | [VERIFIED FROM RAW SIMULATION DATA] | Zero errors in the finite simulated pattern; not a BER guarantee. |

[ENGINEERING INTERPRETATION] These results demonstrate an integrated evaluation workflow and successful candidate snapshots. They do not establish silicon behavior, post-layout performance, PCI-SIG compliance, complete-chip area, or reliable convergence from arbitrary starting points.

### 1.3 Principal limitations

The current work is explicitly Stage 1. The tail source remains ideal, the DFE is behavioral, and the resistors and capacitors are represented by ideal SPICE elements rather than extracted layout devices. The optimizer computes partial matched-MOS channel area, but not passive area, routing, diffusion, contacts, guard rings, sampler area, DFE area, or layout overhead.

[REPORTED IN PPT] The presentation states the same limitations, including behavioral DFE modeling, partial area estimation, incomplete SF/FS coverage, and lack of continuous tuning proof. [VERIFIED FROM SOURCE CODE] The repository documentation makes these boundaries more precise and records a separate 60-condition grid that includes SF and FS. Consequently, the report uses 36/36 only for the documented TT/SS/FF qualification grid and does not describe it as full five-corner qualification.

### 1.4 Requirements-compliance matrix

The matrix below separates implementation from qualification. “Demonstrated” means the repository contains direct source, test, or artifact evidence for the stated scope; it does not imply silicon or protocol compliance.

| Original requirement             | Target/specification       | Current implementation or evidence | Status       | Limitation or qualification                                            |
|----------------------------------|----------------------------|------------------------------------|--------------|------------------------------------------------------------------------|
| 5-GT/s / 2.5-GHz Nyquist context | 5e9 bit/s NRZ; f_N=2.5 GHz | simulator/stimulus.                | Demonstrated | Data-rate context only; not PCI-SIG electrical or protocol compliance. |

| Original requirement | Target/specification                                     | Current implementation or evidence             | Status                                     | Limitation or qualification                                                        |
|----------------------|----------------------------------------------------------|------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------|
| CTLE peaking         | 3–12 dB in intended band                                 | AC metric and hard gate in simulator/receiver. | Demonstrated for evaluated candidates      | Continuous tuning across the full band is not established.                         |
| One-stage CTLE       | Parameterized source-degenerated CTLE                    | circuits/blocks/ctl                            | Demonstrated                               | Stage-1 topology with ideal tail source.                                           |
| Source degeneration  | RDEG    CDEG branches                                    | CTLE netlist and AC bench                      | Demonstrated                               | Passive devices are ideal SPICE elements, not extracted layout devices.            |
| One-tap DFE          | Previous-decision correction in receiver evaluation      | simulator/receiver_                            | Implemented but not fully qualified        | Behavioral model; no physical sampler/DFE signoff.                                 |
| Eye width            | Implemented threshold above 0.4 UI at candidate fidelity | dfe_eye_width_ui metric                        | Demonstrated for selected nominal runs     | Finite pattern and behavioral DFE scope.                                           |
| Eye height           | Implemented threshold above 0.1 V                        | dfe_locked_phase_ey                            | Demonstrated for selected nominal runs     | Repository convention; original external requirement needs confirmation.           |
| HD3                  | Below -30 dB at 100 mVpp differential                    | ctl_hd3.cir, hd3_db()                          | Demonstrated for selected runs             | Applies only to the documented 100 MHz test and amplitude.                         |
| Input-referred noise | Below 1.5 mVrms over implemented band                    | ctl_noise.cir and integration                  | Demonstrated for selected runs             | Applies only to the documented 10 MHz–5 GHz analysis.                              |
| Power                | CTLE power between 0 and 15 mW                           | DC/transient power metrics                     | Demonstrated for CTLE-level accounting     | Total receiver power is unavailable.                                               |
| Area                 | 0.05 mm <sup>2</sup> total-area target                   | Partial matched-MOS channel-area estimate      | Partially demonstrated                     | Passives, routing, contacts, guard rings, sampler, DFE, and overhead are excluded. |
| TT/SS/FF/SF/FS       | Five process corners                                     | Full 60-point grid is defined                  | Implemented but not fully qualified        | SF/FS completion is unavailable.                                                   |
| Supply variation     | 1.71, 1.80, 1.89 V                                       | PVT_GRID and qualification grid                | Demonstrated for historical TT/SS/FF scope | Not evidence of full five-corner qualification.                                    |

| Original requirement | Target/specification                             | Current implementation or evidence                   | Status                                 | Limitation or qualification                                                                      |
|----------------------|--------------------------------------------------|------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------|
| Temperature range    | 0, 27, 75, 125 °C                                | PVT_GRID                                             | Demonstrated for historical grid scope | Fresh final-fidelity completion is unavailable.                                                  |
| Near-optimality      | Prove candidate is near the best feasible design | PPO/RS/CEM comparisons                               | Not established                        | No exhaustive optimum, bound, or statistically powered ranking exists.                           |
| Runtime reduction    | Demonstrate lower cost than exhaustive search    | Caching, staging, early rejection, measured runtimes | Partially demonstrated                 | No controlled end-to-end reduction factor versus an exhaustive baseline.                         |
| Human intervention   | Eliminate manual engineering decisions           | UI, wrapper, checkpoints, reports                    | Not established                        | Targets, channel, model, validation scope, and interpretation still require engineering choices. |

## 2. Project Objective and Design Requirements

### 2.1 Design objective

The objective is to automate the search for a circuit parameter vector

$$\theta = (R_{LOAD}, R_{DEG}, C_{DEG}, I_{TAIL}, V_{DFE}, W_{MOS}, L_{MOS}, M_{MOS})$$

that satisfies receiver-level and circuit-level constraints under a selected simulation condition. Here,  $R_{LOAD}$  and  $R_{DEG}$  are resistances in ohms,  $C_{DEG}$  is capacitance in farads,  $I_{TAIL}$  is tail current in amperes,  $V_{DFE}$  is the behavioral tap voltage in volts,  $W_{MOS}$  and  $L_{MOS}$  are transistor dimensions in micrometers, and  $M_{MOS}$  is an integer multiplier. The first two PPO generations expose only the first five entries; v3 exposes all eight.

[VERIFIED FROM SOURCE CODE] The target specification is represented by four primary quantities: minimum eye height, minimum eye width, minimum decision margin, and maximum CTLE power. Additional hard checks include AC peaking, HD3, noise, and zero observed DFE decision errors when those stages are run.

The optimization problem is therefore constrained rather than a free-form regression problem. A candidate is acceptable only if the relevant metrics  $m_i(\theta, c)$  satisfy their direction-specific limits under condition  $c$ :

- $m_{eye,height}(\theta, c) \geq T_{eye,height}$
- $m_{eye,width}(\theta, c) \geq T_{eye,width}$
- $m_{margin}(\theta, c) \geq T_{margin}$
- $m_{power}(\theta, c) \leq T_{power}$

The actual threshold values are read from the target and project contracts; they must not be inferred from a plot or from a presentation screenshot. Where a run did not measure a metric, the report status is NOT CLAIMED.

### 2.2 Receiver-level requirements

| Requirement | Meaning                                                | Evidence status                                                                         |
|-------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Eye height  | Vertical opening of the sampled receiver eye, in volts | [VERIFIED FROM SOURCE CODE]<br>Metric is computed by <code>receiver_metrics.py</code> . |

| Requirement          | Meaning                                                                         | Evidence status                                                                                 |
|----------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Eye width            | Horizontal opening normalized to one unit interval                              | [VERIFIED FROM SOURCE CODE]<br>Metric is computed from sampled phases.                          |
| Decision margin      | Minimum separation used by the DFE decision process                             | [VERIFIED FROM SOURCE CODE]<br>Metric is part of the receiver result contract.                  |
| CTLE power           | Reported CTLE supply power, in watts                                            | [VERIFIED FROM SOURCE CODE]<br>DC stage computes and gates it.                                  |
| HD3                  | Third-harmonic distortion relative to the fundamental, in dB                    | [VERIFIED FROM SOURCE CODE]<br>Measured by the HD3 bench when enabled.                          |
| Input-referred noise | Integrated input-referred noise, in Vrms                                        | [VERIFIED FROM SOURCE CODE]<br>Measured by the noise bench when enabled.                        |
| PVT robustness       | Repeated qualification over defined corner, voltage, and temperature conditions | [VERIFIED FROM LOGS]<br>Demonstrated for the documented 36-point grid in one historical result. |
| Total area           | Complete physical receiver area                                                 | [NOT AVAILABLE] Only partial MOS channel area exists.                                           |

## 2.3 Requirements that remain conditional or unavailable

[ENGINEERING INTERPRETATION] Eye metrics measured after a behavioral DFE are useful for evaluating the algorithmic receiver contract, but they do not by themselves demonstrate that a physical sampler, clocking circuit, or DFE can implement the same correction. Similarly, a CTLE power value is not total receiver power.

The supplied evidence does not contain a post-layout extraction, a DRC-clean layout, silicon measurements, or commercial-SPICE cross-validation. It also does not establish continuous compliance over every frequency in the stated tuning interval. These items are therefore marked [NOT AVAILABLE] rather than treated as implied by successful nominal simulations.

## 2.4 Figure and table plan for this chapter

Recommended figures are a requirements-to-metrics traceability diagram and a constraint-direction diagram showing lower-bound and upper-bound metrics. The figure should be generated from `rl/target_spec.py`, `simulator/receiver.py`, and `analysis/final_specification.py`; no threshold should be added from the PPT unless it is also present in source code.

# 3. Evidence Classification

## 3.1 Evidence hierarchy

This report uses a deliberately conservative hierarchy. Source code defines what the system is intended to do; raw simulation artifacts show what a particular run produced; logs establish execution and timing; documentation explains historical decisions; and the PPT supplies presentation context. These sources are complementary but are not interchangeable.

| Label                       | Use in this report                              | Example                                                     |
|-----------------------------|-------------------------------------------------|-------------------------------------------------------------|
| [VERIFIED FROM SOURCE CODE] | Directly confirmed from implementation or tests | PPO action deltas, SPICE stage order, PVT grid construction |

| Label                               | Use in this report                                                             | Example                                             |
|-------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|
| [VERIFIED FROM RAW SIMULATION DATA] | Read from JSON/JSONL or waveform-derived result artifacts                      | Eye height, HD3, noise, candidate pass status       |
| [VERIFIED FROM LOGS]                | Confirmed by command output or run logs                                        | Live test count, runtime, completed evaluations     |
| [REPORTED IN PPT]                   | Present in the uploaded slide deck but not independently sufficient            | Dashboard screenshots or slide-level summary claims |
| [ENGINEERING INTERPRETATION]        | Reasoned explanation of design significance                                    | Why staged rejection reduces expected runtime       |
| [NOT AVAILABLE]                     | No traceable evidence exists in the supplied materials                         | Total receiver area, silicon validation             |
| [REQUIRES FURTHER VERIFICATION]     | Evidence exists but is incomplete, historical, or not independently reproduced | Historical qualification at a different fidelity    |

### 3.2 Fresh, historical, synthetic, and planned evidence

Results are separated into four categories:

1. **Fresh real-SPICE evidence** is identified by its raw result artifact or execution log and includes the current test run and wrapper demonstrations.
2. **Historical real-SPICE evidence** exists in committed result artifacts and documentation but was not independently rerun for this report. The 36/36 qualification result is in this category.
3. **Synthetic evidence** exercises software mechanics without ngspice. It can show that PPO learns a controlled algebraic landscape, but it cannot prove circuit performance.
4. **Planned or missing evidence** identifies work described in plans or limitations but not supported by a completed artifact.

[VERIFIED FROM SOURCE CODE] The synthetic benchmark is explicitly labelled as non-circuit evidence in both its implementation and its tests. This prevents a fast synthetic training curve from being used as a substitute for a real SPICE result.

### 3.3 Reproducibility procedure

A reproducible claim in this report must identify, where available:

- repository revision and branch;
- source netlists and model-library identity;
- channel file, port map, and checksum;
- PPO version, checkpoint metadata, exact parameter grids, and seed;
- simulation fidelity and operating condition;
- target specification and metric thresholds;
- result file or live command that produced the value;
- whether the result was fresh, historical, or synthetic.

[VERIFIED FROM SOURCE CODE] `simulator/provenance.py` hashes implementation and model inputs, and `simulator/ngspice.py` records executable and solver identity. [VERIFIED FROM SOURCE CODE] v3 checkpoints store a metadata envelope because the policy cannot safely be interpreted without its state schema and parameter grid. [ENGINEERING INTERPRETATION] These mechanisms reduce accidental mixing of incompatible results, although they do not replace independent reruns.

### 3.4 Handling discrepancies

The repository contains several result families produced at different times, fidelities, and interfaces. Values are not merged merely because their labels look similar. For example, a nominal eye-width value from a fresh wrapper run is not

substituted into a historical PVT table, and a presentation screenshot is not used to replace a raw JSON artifact.

The PPT was inspected slide by slide, including its embedded images. It is used as [REPORTED IN PPT] context only. When a slide states “36 tested PVT conditions,” this report cross-references QUALIFICATION\_PVT\_GRID and states the precise TT/SS/FF scope. The separate SF/FS corners in the 60-condition grid remain outside the demonstrated qualification claim.

### 3.5 Reproducibility table

| Evidence item            | Repository source                    | Status                              | Reproduction note                                     |
|--------------------------|--------------------------------------|-------------------------------------|-------------------------------------------------------|
| Source revision          | Git branch and HEAD                  | [VERIFIED FROM LOGS]                | r1-ppo-v2-study, HEAD 11101a5.                        |
| Unit/regression tests    | tests/                               | [VERIFIED FROM LOGS]                | 632 passed, 2 skipped, 27 subtests.                   |
| v3 checkpoint            | results/ppo_v3_tt_1000.p             | [VERIFIED FROM RAW SIMULATION DATA] | Metadata-bearing v3 policy accepted by the v3 loader. |
| Historical PVT result    | results/v3_qualification             | [REQUIRES FURTHER VERIFICATION]     | Present and documented, not rerun in this pass.       |
| Synthetic training       | r1/synthetic_benchmark.p and results | [VERIFIED FROM SOURCE CODE]         | Software-mechanics evidence only.                     |
| Whole-chip physical area | No complete artifact                 | [NOT AVAILABLE]                     | Must not be reconstructed from partial channel area.  |

## 4. System Architecture

### 4.1 Architectural layers

Nebula is organized as a layered pipeline rather than a single monolithic optimization script:

1. **Interface layer.** The local web UI and the natural-language wrapper accept targets, backend choices, checkpoint selections, channel options, and validation settings.
2. **Orchestration layer.** `experiments/run_autockt_pipeline.py` validates the target, generates candidates, filters nominal results, optionally runs PVT and final measurements, and writes structured outputs.
3. **Optimization layer.** PPO, Random Search, and CEM produce parameter candidates under their respective contracts.
4. **Evaluation layer.** `simulator.receiver.evaluate_receiver()` performs staged DC, AC, transient, channel, noise, HD3, and receiver analyses.
5. **Analysis layer.** Specification assessment, PVT selection, comparison reports, plots, schematic export, and dashboard data consume structured results rather than re-running simulations.

[VERIFIED FROM SOURCE CODE] The web UI intentionally runs the pipeline in a subprocess and does not import PPO, simulator, reward, or PVT-selection logic into the server process. The LLM wrapper also constructs a pipeline command and consumes its JSON result. This gives the interfaces a common source of truth and provides process isolation for long-running or native-SPICE work.

### 4.2 End-to-end data flow

The principal data flow is:

TargetSpec → checkpoint / optimizer →  $\theta$  → ReceiverEvaluation → metric assessment → candidate selection → report and netlist.

The candidate vector  $\theta$  is rendered into SPICE parameters. The receiver evaluation returns stage results, measured metrics, violations, warnings, failure stage, runtime, evaluation identity, and provenance. The pipeline then treats the result as structured data: it does not infer a pass from a missing value and it does not allow an incomplete PVT run to qualify a design.

### 4.3 Failure-aware control flow

Each stage can return success or a structured failure. The receiver stops on a hard failure, records the stage name, and avoids spending later-stage runtime on a candidate that has already violated an earlier requirement. Typical failure labels include `setup`, `dc`, `ac`, `ctle_transient`, `channel`, `transient`, `noise`, `hd3`, and `internal` at the receiver level; the lower `ngspice` contract additionally distinguishes `launch`, `timeout`, `parsing`, `measurement`, and `model-library` failures.

[ENGINEERING INTERPRETATION] Early rejection is both an engineering safety mechanism and a runtime strategy. It prevents a nominally invalid bias point from being presented as a meaningful receiver result, while allowing the optimizer to spend its limited evaluation budget on candidates that reach informative metrics.

### 4.4 Architecture evidence table

| Layer                    | Main implementation                                           | Input                                   | Output                                     | Evidence                    |
|--------------------------|---------------------------------------------------------------|-----------------------------------------|--------------------------------------------|-----------------------------|
| Web interface            | <code>experiments/web_ui.py</code>                            | User configuration                      | Subprocess request and displayed artifacts | [VERIFIED FROM SOURCE CODE] |
| Natural-language wrapper | <code>nebula/llm_wrapper.py</code>                            | Text target/request                     | Validated pipeline arguments               | [VERIFIED FROM SOURCE CODE] |
| Pipeline                 | <code>experiments/run_autopilot.py</code>                     | Target, checkpoint, backend, conditions | Candidate and qualification JSON           | [VERIFIED FROM SOURCE CODE] |
| PPO environment          | <code>rl/autockt_env.py</code> and <code>rl/trainer.py</code> | State, action, evaluator result         | Transition and reward                      | [VERIFIED FROM SOURCE CODE] |
| Receiver evaluator       | <code>simulator/receiver.py</code>                            | Parameters and simulation conditions    | <code>ReceiverEvaluation</code>            | [VERIFIED FROM SOURCE CODE] |
| ngspice adapter          | <code>simulator/ngspice.py</code>                             | Rendered netlist and model library      | Typed simulation result                    | [VERIFIED FROM SOURCE CODE] |
| Analysis/reporting       | <code>analysis/</code>                                        | Structured result artifacts             | Verdicts, plots, summaries, exports        | [VERIFIED FROM SOURCE CODE] |

### 4.5 Candidate selection and qualification boundary

Candidate generation and qualification are deliberately separate. PPO may produce a candidate at a cheaper training fidelity, but the pipeline can re-evaluate selected designs at candidate or final fidelity and then apply a PVT condition set. A candidate that passes nominal metrics is therefore not automatically PVT-qualified.

The output report contains PASS, FAIL, and NOT CLAIMED rows. A PVT row is only promoted when the required condition set is actually evaluated and the qualification logic accepts it. Incomplete or failed checks remain visible as limitations rather than being silently omitted.

## 5. PCIe Gen-2 Receiver and Channel Model

### 5.1 Signaling context

[VERIFIED FROM SOURCE CODE] `simulator/stimulus.py` defines the receiver stimulus bit rate as 5e9 bits/s. For binary NRZ signaling, the unit interval is the time occupied by one bit:

$$UI = \frac{1}{R_b}$$

where  $R_b$  is the bit rate in bits/s and  $UI$  is in seconds. The associated Nyquist frequency is  $f_N = R_b/2$ . For the implemented 5 Gbit/s setting,  $f_N$  is 2.5 GHz by this definition. This is the frequency region in which a lossy channel and an equalizer must preserve enough signal separation for the receiver's eye metrics to remain positive.

[ENGINEERING INTERPRETATION] The report uses "PCIe Gen-2-rate" to describe the implemented 5 GT/s stimulus context, not to claim protocol compliance. A complete PCIe receiver claim would also require protocol, package, clocking, jitter, calibration, and compliance-test evidence. Such evidence is not available in the provided evidence.

## 5.2 Differential signal path

The implemented signal path is:

$v_{TX,d}(t) \rightarrow H_{CH,d}(f) \rightarrow v_{IN,d}(t) \rightarrow CTLE \rightarrow v_{OUT,d}(t) \rightarrow \text{sampling and DFE} \rightarrow b_{\hat{k}}$ .

Here  $v_{TX,d}(t)$  is the transmitted differential waveform in volts,  $H_{CH,d}(f)$  is the channel's differential transfer function,  $v_{IN,d}(t)$  is the filtered receiver input,  $v_{OUT,d}(t)$  is the CTLE differential output, and  $b_{\hat{k}}$  is the binary decision for symbol  $k$ .

[VERIFIED FROM SOURCE CODE] `simulator.receiver._run_transient()` first creates a deterministic NRZ stimulus, filters its differential waveform through the loaded S4P channel, writes the resulting waveform into a temporary SPICE stimulus include, and then evaluates `receiver_transient.cir`. The bench includes 50-ohm source resistance per leg, a differential receiver termination, optional input capacitance, and the CTLE load capacitance from the simulation conditions.

## 5.3 Touchstone S-parameter representation

[VERIFIED FROM SOURCE CODE] `simulator.channel.load_s4p()` reads a Touchstone 1.x four-port file. Each frequency row contains the frequency followed by the 16 complex S-parameters in full-matrix row-major order. The parser supports RI, magnitude-angle, and dB-angle formats and requires a declared reference impedance. The parsed matrix is represented as

$$\mathbf{S}(f) = [S_{ij}(f)]_{i,j=1}^4$$

where  $S_{ij}$  is the complex scattering response at frequency  $f$  from source port  $j$  to destination port  $i$ . Frequencies must be strictly increasing, and the parser rejects truncated or unsupported data.

The differential transfer is formed from the selected transmit and receive ports as

$$S_{dd21}(f) = \frac{1}{2}(S_{r^+t^+} - S_{r^+t^-} - S_{r^-t^+} + S_{r^-t^-})$$

The symbols  $t^+$  and  $t^-$  identify the positive and negative transmitter ports, while  $r^+$  and  $r^-$  identify the corresponding receiver ports. The factor  $1/2$  is part of the implemented mixed-mode combination.

[VERIFIED FROM SOURCE CODE] The port map is explicit and validated to use ports 1, 2, 3, and 4 exactly once. The IEEE/IBM metadata selects physical ports 1 and 3 for the transmitter pair and 2 and 4 for the receiver pair, because its documented dominant forward paths are  $S_{21}$  and  $S_{43}$ . The synthetic fixture uses the repository's default mapping unless another map is supplied.

## 5.4 Channel loss and delay

The channel insertion loss reported by Nebula is

$$L_{CH}(f) = 20\log_{10} |S_{dd21}(f)| \quad \text{dB}$$

The implementation interpolates the complex transfer by interpolating magnitude and unwrapped phase separately. [VERIFIED FROM SOURCE CODE] `s4PChannel.insertion_loss_db()` applies this calculation at requested frequencies, and the validation path evaluates 1.25 GHz, 2.5 GHz, and 5 GHz.

Bulk delay is estimated from a least-squares line fit to unwrapped phase:

$$\phi(f) \approx af + b, \quad \tau_{bulk} = -\frac{a}{2\pi}$$

where  $\phi$  is phase in radians,  $f$  is frequency in hertz,  $a$  is the phase slope in radians per hertz, and  $\tau_{bulk}$  is delay in seconds. [VERIFIED FROM SOURCE CODE] The fit uses the 100 MHz to 5 GHz interval and rejects a material negative delay.

The integer delay alignment is

$$n_{delay} = \text{round} \left( \frac{\tau_{bulk}}{UI} \right)$$

implemented with a round-half-up rule that stabilizes floating-point ties. The remaining fractional phase is retained as the sampling reference phase.

## 5.5 Channel validation gates

[VERIFIED FROM SOURCE CODE] `validate_s4p_channel()` applies deterministic preflight checks before receiver transient evaluation.

| Check               | Implementation requirement                      | Unit          | Evidence                    |
|---------------------|-------------------------------------------------|---------------|-----------------------------|
| Reference impedance | Exactly 50 within the implementation tolerance  | ohm           | [VERIFIED FROM SOURCE CODE] |
| Frequency coverage  | Data extends through at least 5e9               | Hz            | [VERIFIED FROM SOURCE CODE] |
| Finite transfer     | All differential-transfer values finite         | —             | [VERIFIED FROM SOURCE CODE] |
| Passivity screen    | Maximum singular value no greater than 1.01     | dimensionless | [VERIFIED FROM SOURCE CODE] |
| Delay               | No material negative fitted bulk delay          | s             | [VERIFIED FROM SOURCE CODE] |
| Causality screen    | Negative-time energy ratio no greater than 0.05 | dimensionless | [VERIFIED FROM SOURCE CODE] |

The passivity diagnostic computes the largest singular value of the full S4P matrix over the validation band. The causality diagnostic removes the fitted transport delay, performs a band-limited inverse real FFT, and compares the energy in the wrapped negative-time half with total impulse-response energy. [ENGINEERING INTERPRETATION] These are deterministic integration gates, not a formal rational-model or VNA signoff. The source code explicitly describes them as a qualification gate and recommends trusted external modelling tools for final signoff.

### 5.6 Available channel evidence

| Channel                   | Role                              | Evidence status                     | Limitation                                              |
|---------------------------|-----------------------------------|-------------------------------------|---------------------------------------------------------|
| synthetic_regression.s4p  | Deterministic software regression | [VERIFIED FROM DOCUMENTATION]       | Synthetic and not circuit evidence.                     |
| ieee802_ibm_20db_thru.s4p | Public lossy reference channel    | [VERIFIED FROM RAW SIMULATION DATA] | Its metadata states it is not PCIe compliance evidence. |

[REPORTED IN PPT] The presentation shows channel and receiver-flow material. [VERIFIED FROM DOCUMENTATION] The repository sidecar is the controlling source for channel identity, provenance, port assignment, and compliance limitations.

## 6. CTLE Circuit Topology

### 6.1 Differential-pair structure

[VERIFIED FROM SOURCE CODE] circuits/blocks/ctle.spice defines the reusable CTLE subcircuit. It contains two equal load resistors from vdd to outp and outn, two matched sky130\_fd\_pr\_\_nfet\_01v8 devices, two identical source-degeneration branches, and an ideal DC tail-current source.

The differential input and output variables are

$$v_{in,d} = v(inp) - v(inn), \qquad v_{out,d} = v(outn) - v(outp)$$

The reversed output order is deliberate. [VERIFIED FROM SOURCE CODE] The AC, transient, and noise benches comment that the common-source pair inverts the signal and therefore uses outn-outp as the logical receiver polarity.

### 6.2 Implemented circuit parameters

| Parameter | Meaning                             | Range/value                                          | Source file                       | Engineering role                                                                                         |
|-----------|-------------------------------------|------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------|
| RLOAD     | Each drain load resistance          | Baseline 1 kΩ;<br>automated legacy range 100 Ω–10 kΩ | README.md,<br>simulator/design_sc | Converts drain current variation into output voltage; trades gain against bandwidth and operating point. |
| RDEG      | Each source-degeneration resistance | Baseline 1 kΩ;<br>automated legacy range 10 Ω–10 kΩ  | Same                              | Reduces local feedback-dependent gain and improves linearity; shapes the low-frequency degeneration.     |

| Parameter | Meaning                                          | Range/value                                           | Source file         | Engineering role                                                                                      |
|-----------|--------------------------------------------------|-------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------|
| CDEG      | Capacitor parallel to each degeneration resistor | Baseline 0.5 pF; automated legacy range 10 fF–10 pF   | Same                | Reduces degeneration at higher frequency and creates the frequency-dependent equalization transition. |
| ITAIL_VAL | Ideal differential-pair tail current             | Baseline 100 $\mu$ A; automated range 10 $\mu$ A–1 mA | Same                | Sets bias current, transconductance, speed, and CTLE power.                                           |
| DFE_TAP   | Behavioral feedback tap voltage                  | -0.4 V to 0.4 V                                       | ReceiverParameters. | Compensates the previous-symbol contribution in the software DFE.                                     |
| MOS_W     | Matched NMOS width                               | v3 grid/bounds 0.42–100 $\mu$ m                       | simulator/design_sc | Changes drive strength, capacitance, and partial channel area.                                        |
| MOS_L     | Matched NMOS channel length                      | v3 grid/bounds 0.15–1.0 $\mu$ m                       | Same                | Changes device electrical behavior and partial channel area.                                          |
| MOS_M     | Matched device multiplicity                      | Integer 1–16                                          | Same                | Represents parallel device instances and affects drive and partial area.                              |

The baseline values above are [VERIFIED FROM DOCUMENTATION] values recorded by the repository. The automated ranges and validation bounds are [VERIFIED FROM SOURCE CODE]. A range is not a manufacturability claim: the project documentation explicitly says the v3 grids are bounded sizing grids, not proof of DRC closure.

### 6.3 Small-signal interpretation

For a symmetric differential pair, a simplified small-signal view gives an output differential gain of the form

$$A_{vd}(s) \approx -g_m(s) Z_{load}(s) A_{deg}(s)$$

where  $g_m$  is effective transconductance in siemens,  $Z_{load}$  is the frequency-dependent load impedance in ohms, and  $A_{deg}$  represents the source-degeneration feedback factor. This equation is an engineering model, not the exact SKY130 compact-model equation used by ngspice.

[ENGINEERING INTERPRETATION] Increasing  $R_{LOAD}$  can increase low-frequency voltage gain but also increases the output-node time constant with parasitic and explicit load capacitance. Increasing  $R_{DEG}$  strengthens local feedback, which generally lowers gain and improves linearity.  $C_{DEG}$  bypasses part of that degeneration at higher frequency, allowing a controlled transition from lower low-frequency gain to higher high-frequency gain. The exact result is determined by the transistor model, bias point, loads, and channel, so these directions should not be read as monotonic guarantees for every candidate.

### 6.4 Source degeneration and peaking

The two branches are implemented as

$$Z_{deg}(s) = R_{DEG} \parallel \frac{1}{sC_{DEG}} = \frac{R_{DEG}}{1 + sR_{DEG}C_{DEG}}$$

Here  $s=j2\pi f$ ,  $f$  is frequency in hertz,  $R_{DEG}$  is ohms, and  $C_{DEG}$  is farads. The characteristic transition frequency of this simple parallel network is

$$f_{deg} = \frac{1}{2\pi R_{DEG}C_{DEG}}$$

[ENGINEERING INTERPRETATION] This equation explains why the R/C pair is a natural CTLE control: below the transition, the resistor contributes strong degeneration; above it, the capacitor lowers the effective degeneration impedance. The actual response can include additional poles and zeros from the MOS devices, loads, and wiring represented by the model.

The project does not use “peaking” as an informal visual judgement. It extracts gain from the AC trace and computes the difference between the peak gain in the intended 1.25–2.5 GHz band and the gain at 100 MHz:

$$P_{CTLE} = G_{peak, 1.25-2.5 \text{ GHz}} - G_{100 \text{ MHz}} \quad \text{dB}$$

[VERIFIED FROM SOURCE CODE] `simulator.waveform.ac_metrics()` also classifies the response as a local peak, controlled shelf, or uncontrolled rise and checks the out-of-band peak and shelf behavior.

## 6.5 Tail current and MOS sizing

[VERIFIED FROM SOURCE CODE] The CTLE subcircuit uses an ideal source `ITAIL tail vss DC {ITAIL_VAL}`. The implementation does not contain a transistor-level bias generator in this Stage-1 block.

[ENGINEERING INTERPRETATION] For a fixed topology, `I_TAIL` changes the available branch current and therefore the MOS operating point and effective  $g_m$ . Higher current can improve speed or linearity but increases static power. The actual trade-off is evaluated by ngspice rather than assumed from a hand-derived square-law model.

The matched sizing group is represented by one width, one length, and one multiplier shared by both devices. The partial channel-area expression used by v3 is

$$A_{MOS, partial} = 2 W_{MOS} L_{MOS} M_{MOS}$$

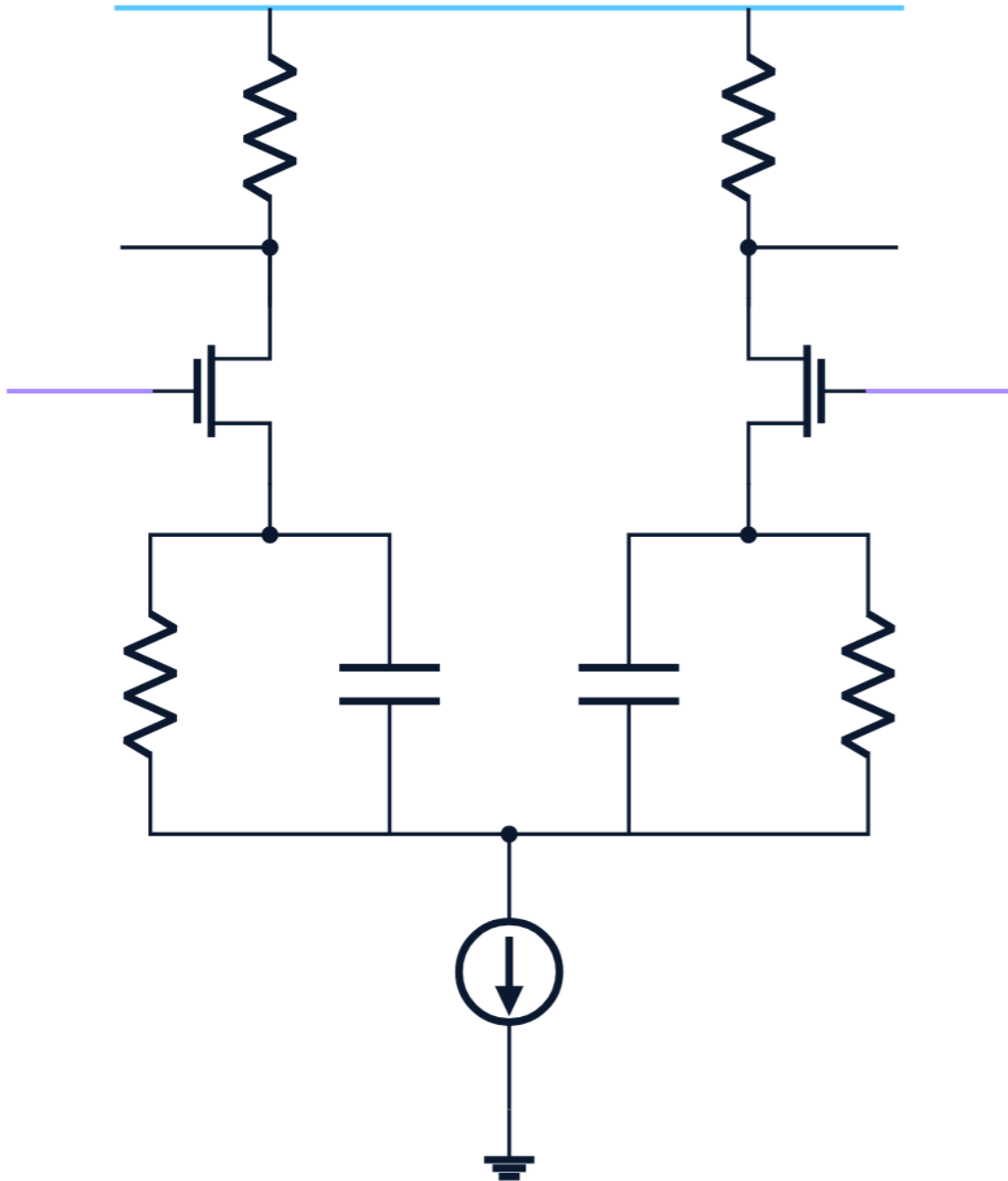
where the factor 2 represents the two matched transistors and the dimensions are expressed consistently in the area-reporting calculation. [VERIFIED FROM SOURCE CODE] This quantity is explicitly partial and is never promoted to a total-area `PASS`.

## 6.6 Circuit and simulation trade-offs

| Design change               | Intended benefit                           | Possible cost                              | Evidence boundary                                                   |
|-----------------------------|--------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|
| Increase <code>RLOAD</code> | More voltage conversion from drain current | Lower bandwidth or altered output headroom | [ENGINEERING INTERPRETATION] Exact behavior is simulator-dependent. |

| Design change           | Intended benefit                                           | Possible cost                      | Evidence boundary            |
|-------------------------|------------------------------------------------------------|------------------------------------|------------------------------|
| Increase RDEG           | More degeneration and potentially improved linearity       | Lower gain and altered peaking     | [ENGINEERING INTERPRETATION] |
| Increase CDEG           | Earlier high-frequency degeneration bypass                 | Excessive peaking or noise shaping | [ENGINEERING INTERPRETATION] |
| Increase ITAIL_VAL      | More bias current and possible speed/linearity improvement | More CTLE power                    | [ENGINEERING INTERPRETATION] |
| Increase MOS_W or MOS_M | More drive and transconductance opportunity                | More capacitance and partial area  | [ENGINEERING INTERPRETATION] |
| Increase MOS_L          | Changes current and intrinsic device behavior              | Bandwidth and area trade-offs      | [ENGINEERING INTERPRETATION] |

The table describes engineering rationale, not guaranteed one-dimensional monotonicity. [REQUIRES FURTHER VERIFICATION] A complete sensitivity study with matched real-SPICE perturbations for every geometry and passive variable is not available in the provided evidence.



**Figure 1.** Source-degenerated differential CTLE schematic supplied in the Nebula presentation. It shows the matched MOS pair, resistive loads, parallel degeneration R/C branches, and ideal tail-current source. **Evidence:** `report_generation/figures/ppt_ctle_topology.png`, extracted from the uploaded PPT; [REPORTED IN PPT]. The netlist in `circuits/blocks/ctle.spice` remains the primary implementation evidence.

## 7. Behavioral One-Tap DFE

### 7.1 Purpose and model boundary

[VERIFIED FROM SOURCE CODE] `simulator/receiver_metrics.py` implements a stateful one-tap DFE. The DFE operates on sampled CTLE output values after the channel and does not instantiate a transistor-level sampler, latch, feedback amplifier, or physical decision circuit.

The DFE's purpose is to remove an estimated contribution from the preceding symbol. Let  $x[k]$  be the raw sampled voltage in volts,  $d[k-1]$  be the previous binary decision represented as a sign in  $\{-1,+1\}$ , and  $V\_T$  be the decision threshold in volts. The implemented correction is

$$y[k] = x[k] - V_{tap} d[k - 1]$$

followed by

$b\_hat[k] = 1$  when  $y[k] \geq V\_T$ , and  $b\_hat[k] = 0$  when  $y[k] < V\_T$ .

Here  $V\_tap$  is `dfe_tap_v`,  $y[k]$  is the corrected sample, and  $b\_hat[k]$  is the binary decision. [VERIFIED FROM SOURCE CODE] The default threshold is zero volts and the initial previous bit is validated to be either zero or one.

## 7.2 Training and decision-directed operation

Nebula uses a hybrid mode. During the known training region, the expected bit is used to update the previous-symbol feedback state. After `training_stop`, the DFE uses its own decision, creating decision-directed feedback.

If  $b[k]$  is the expected bit, its sign is

$s[k] = +1$  when  $b[k] = 1$ , and  $s[k] = -1$  when  $b[k] = 0$ .

The signed margin recorded by the implementation is

$$m[k] = s[k](y[k] - V_T) \quad V$$

Positive  $m[k]$  means the corrected sample is on the expected side of the threshold. [VERIFIED FROM SOURCE CODE]

The DFE result records corrected samples, decisions, expected bits, margins, error indices, error count, and error-propagation events.

## 7.3 Phase selection

The receiver does not assume that the center of the nominal UI is optimal after channel delay and CTLE filtering.

`choose_sampling_phase()` evaluates a grid of phases from 0 to one UI using the transient time step. For each phase, it samples the waveform, applies the DFE while all phase-training feedback symbols are known, and scores the phase using the tenth percentile of the training-region signed margins:

$$S(\phi) = \text{percentile}_{10\%} \{m[k; \phi]\}_{k \in T}$$

Here  $\phi$  is phase in seconds,  $T$  is the phase-training index set, and  $S(\phi)$  is in volts. [VERIFIED FROM SOURCE CODE]

The selected phase maximizes  $S(\phi)$ . Plateaus are resolved using the longest contiguous best run and a reference phase derived from the channel delay.

[ENGINEERING INTERPRETATION] Using a lower-tail percentile rather than only a mean margin makes phase selection sensitive to the weaker part of the training distribution. It is still a finite-pattern heuristic, not an eye-opening theorem or a clock-recovery implementation.

## 7.4 Eye height and eye width

For each candidate phase, the implementation separates corrected samples belonging to logical ones and zeros. The vertical DFE eye height is computed from the tenth percentile of the one samples and the ninetieth percentile of the zero samples:

$$H_{eye}(\phi) = P_{10}(y \mid b = 1, \phi) - P_{90}(y \mid b = 0, \phi) \quad V$$

The best phase is the phase with the largest finite  $H_{eye}$ . A phase is considered open when its height is at least the configured minimum height. The implementation then finds the contiguous open region containing the locked phase and converts its number of phase samples to unit intervals:

$$W_{eye, UI} = N_{open} \frac{\Delta t_{phase}}{UI}$$

Here  $N_{open}$  is the contiguous sample count and  $\Delta t_{phase}$  is the phase-step duration in seconds. [VERIFIED FROM SOURCE CODE] At candidate fidelity and above, the receiver requires locked-phase eye height above 0.1 V and eye width above 0.4 UI. Those thresholds are implementation gates, not a claim that they are a complete PCIe compliance specification.

## 7.5 Error count and decision margin

For the measured region  $M$ , the implementation records

$$N_{err} = \sum_{k \in M} \mathbf{1}[\hat{b}[k] \neq b[k]]$$

and

$$E_{err} = \frac{N_{err}}{|M|}$$

The reported minimum decision margin is

$$M_{min} = \min_{k \in M} m[k] \quad \text{V}$$

[VERIFIED FROM SOURCE CODE] The transient stage rejects a candidate if the post-warm-up error count is nonzero or if the minimum margin is not positive. Zero observed errors means only that the finite simulated pattern contained no observed decision errors under the tested condition. [ENGINEERING INTERPRETATION] It is not a measured bit-error rate at an extrapolated target confidence and does not prove long-run or silicon BER.

## 7.6 Behavioral-model assumptions and limitations

The model assumes a known deterministic stimulus, an idealized sampling operation, a scalar one-tap correction, a fixed threshold, and a feedback state that switches from known training bits to hard decisions. It does not model transistor

noise, clock jitter, sampler kickback, comparator offset, metastability, finite feedback delay, adaptation dynamics, mismatch, or physical DFE power and area.

[VERIFIED FROM SOURCE CODE] The generated eye artifact labels the CTLE trace as a continuous transistor-level CTLE output and the DFE points as behavioral one-tap corrected samples; it explicitly states that the DFE plot is not a transistor-level DFE waveform. [REPORTED IN PPT] The presentation shows the behavioral DFE boundary as a project limitation. A transistor-level DFE implementation is not available in the provided evidence.

## 8. SPICE Benchmarks and Simulation Metrics

### 8.1 Netlist templating and model use

[VERIFIED FROM SOURCE CODE] Each bench is a parameterized ngspice template.

`simulator.ngspice.run_simulation()` renders placeholders for the model library, process corner, temperature, CTLE block path, stimulus include, and output file. The request carries typed parameters and expected output artifacts. A temporary run directory isolates generated files, and the result contract records status, measurements, runtime, errors, return code, failure stage, failure code, artifacts, and provenance.

The shared CTLE bench parameters include supply voltage  $V_{DD}$  in volts, input common-mode voltage  $V_{CM}$  in volts, output load capacitance  $C_{LOAD}$  in farads, source resistance per leg in ohms, differential termination in ohms, and input parasitic capacitance in farads. Exact defaults are read from `SimulationConditions` and the bench templates; a value not present in the relevant artifact is not inferred here.

### 8.2 DC operating-point benchmark

`ctle_dc.cir` performs an `.op` analysis and prints input, output, source, tail, and supply-current quantities. [VERIFIED FROM SOURCE CODE] The receiver uses these measurements to form DC metrics and checks output offset, branch current balance, output common-mode behavior, rail headroom, and CTLE power.

The DC power relationship represented by the evaluator is

$$P_{DC} = -V_{DD} I_{VDD} \quad W$$

where  $I_{VDD}$  is the ngspice supply-current sign convention and the leading minus sign converts the source current convention into consumed power.

[ENGINEERING INTERPRETATION] DC screening is placed first because a failed bias point makes later AC and transient metrics difficult to interpret. The DC result is also not total receiver power: the Stage-1 circuit contains an ideal tail source and does not include a physical sampler or physical DFE.

### 8.3 AC transfer benchmark

`ctle_ac.cir` applies a differential small-signal excitation with opposite AC polarity on the two inputs, sweeps logarithmically from 1 MHz to 20 GHz, and writes transfer magnitude, phase, input magnitude, and output magnitude. [VERIFIED FROM SOURCE CODE] The logical differential transfer is

$$H_{CTLE}(f) = \frac{V_{out,d}(f)}{V_{in,d}(f)}$$

and gain in decibels is

$$G(f) = 20\log_{10} |H_{CTLE}(f)| \quad \text{dB}$$

The parser interpolates gain at 1 MHz, 10 MHz, 100 MHz, 1.25 GHz, 2.5 GHz, and 5 GHz. It identifies the peak in the intended 1.25–2.5 GHz band and computes CTLE peaking relative to 100 MHz. It additionally checks whether an out-of-band peak is excessive and whether the 2.5–5 GHz shelf is controlled.

The phase trace is unwrapped and converted to group delay using

$$\tau_g(f) = -\frac{d\phi(f)}{d(2\pi f)} \quad \text{S}$$

The implementation records mean and span over 100 MHz to 5 GHz and emits a warning if the span exceeds the documented implementation threshold.

#### 8.4 CTLE transient diagnostic

`ctle_transient.cir` uses a channel-free terminated differential waveform to check that the CTLE produces a nonzero output swing and receives the expected terminated input amplitude. [VERIFIED FROM SOURCE CODE] The evaluator parses `vin_diff` and `vout_diff`, verifies the input peak-to-peak excursion, and rejects zero output swing.

This is a diagnostic rather than the final receiver metric. It isolates the CTLE and its termination from channel filtering and DFE sampling, allowing a failure to be assigned to the local circuit before a more expensive channel transient is attempted.

#### 8.5 Channel-driven transient benchmark

`receiver_transient.cir` writes the input differential voltage, output differential voltage, both output nodes, and supply current. The evaluator filters the generated NRZ waveform through the channel before writing the stimulus include. It then aligns the waveform by integer bulk delay, selects a fractional sampling phase, applies the behavioral DFE, and computes eye, margin, error, delay, loss, and average transient-power metrics.

[VERIFIED FROM SOURCE CODE] The transient configuration changes by fidelity: SCREENING uses 32 bits, TRAINING uses 128 bits, CANDIDATE uses 512 bits, and FINAL uses 1024 bits. Warm-up and tail symbols are excluded from the measured region. The measured bit count can therefore be lower than the requested stimulus count after warm-up, tail, and channel-delay removal.

The average transient power is computed as

$$P_{avg} = -V_{DD} \text{mean}(I_{VDD}(t)) \quad \text{W}$$

over the valid measurement interval. Output rail excursions are also checked against the nominal supply and ground with the implementation's interpolation tolerance.

#### 8.6 Input-referred noise benchmark

`ctle_noise.cir` runs ngspice noise analysis with the differential output observation and a differential test source. The output includes input-noise spectral density and output-noise spectral density. [VERIFIED FROM SOURCE CODE] `integrated_input_noise()` integrates the square of the input-referred density over frequency:

$$v_{n, rms} = \sqrt{\int_{f_1}^{f_2} S_{v, in}^2(f) df} \quad V_{rms}$$

where  $S_{v, in}(f)$  is input-referred noise density in  $V/\sqrt{Hz}$ ,  $f_1$  and  $f_2$  are the analysis limits in hertz, and trapezoidal numerical integration is used. The implemented bench spans 10 MHz to 5 GHz. The noise stage rejects values at or above 1.5 mVrms.

[ENGINEERING INTERPRETATION] This is an integrated small-signal noise result for the modeled CTLE and test source. It does not include unmodeled physical DFE, clock, package, or post-layout noise contributions.

## 8.7 HD3 benchmark

`ctle_hd3.cir` applies a differential sinusoidal input and records the output differential waveform. The bench uses a 100 MHz fundamental and a differential peak-to-peak amplitude supplied by the evaluator. [VERIFIED FROM SOURCE CODE] At candidate fidelity one amplitude is used; at final characterization the implementation evaluates 0.05 V, 0.1 V, and 0.2 V differential peak-to-peak amplitudes. The reported gate metric is the 0.1 V case.

After discarding the first 100 ns, `hd3_db()` fits the output to fundamental, second-harmonic, third-harmonic, DC, and slow-drift basis functions. If  $A_1$  and  $A_3$  are the fitted fundamental and third-harmonic amplitudes in volts, the reported distortion is

$$HD3 = 20 \log_{10} \left( \frac{A_3}{A_1} \right) \quad \text{dB}$$

The implementation requires steady-state duration, sufficient Nyquist margin, and a well-conditioned fit. The gate requires the 100 mVpp result to be below -30 dB.

[ENGINEERING INTERPRETATION] HD3 measures one specified sinusoidal operating condition. It is not a broadband linearity proof, and the documented bench frequency is not a sweep across the CTLE peaking band. [REQUIRES FURTHER VERIFICATION] A full-amplitude, full-band distortion characterization is not available in the provided evidence.

## 8.8 Metric extraction and failure semantics

| Analysis       | Raw output                     | Extracted metric                             | Failure examples                                                      | Evidence                    |
|----------------|--------------------------------|----------------------------------------------|-----------------------------------------------------------------------|-----------------------------|
| DC             | Printed operating-point values | Bias, current, common-mode, power            | Missing model, invalid bias, current imbalance                        | [VERIFIED FROM SOURCE CODE] |
| AC             | wrdata transfer trace          | Gain, phase, peaking, shelf, group delay     | Missing trace, invalid magnitude, uncontrolled response               | [VERIFIED FROM SOURCE CODE] |
| CTLE transient | vin_diff, vout_diff            | Input/output swing                           | Small input or zero output                                            | [VERIFIED FROM SOURCE CODE] |
| Channel        | Touchstone matrix              | Loss, delay, singular value, causality ratio | Wrong impedance, insufficient bandwidth, non-passive/noncausal screen | [VERIFIED FROM SOURCE CODE] |

| Analysis           | Raw output                      | Extracted metric                          | Failure examples                                              | Evidence                    |
|--------------------|---------------------------------|-------------------------------------------|---------------------------------------------------------------|-----------------------------|
| Receiver transient | Output nodes and supply current | Eye, width, margin, errors, average power | Error, nonpositive margin, rail excursion, insufficient eye   | [VERIFIED FROM SOURCE CODE] |
| Noise              | Input/output noise spectra      | Integrated input-referred Vrms            | Missing trace or threshold violation                          | [VERIFIED FROM SOURCE CODE] |
| HD3                | Differential transient          | Third/fundamental ratio                   | Insufficient cycles, ill-conditioned fit, threshold violation | [VERIFIED FROM SOURCE CODE] |

## 8.9 Reproducible benchmark flow

The benchmark flow is deterministic at the software level when its parameter vector, model library, simulator, channel, stimulus, fidelity, and conditions are held constant. [VERIFIED FROM SOURCE CODE] The evaluator records checksums for the CTLE block, all bench files, implementation modules, model-library dependencies, channel, and simulator identity. Temporary waveform artifacts are parsed into scalar metrics and normally discarded, except when an explicit eye capture is requested.

The exact raw AC, noise, and HD3 traces for every historical result are not all retained as standalone report-ready plots. [NOT AVAILABLE IN THE PROVIDED EVIDENCE] Where only scalar result JSON is present, this report does not invent a waveform plot.

## 9. Hierarchical Evaluation Pipeline

### 9.1 Evaluation contract

[VERIFIED FROM SOURCE CODE] `simulator.receiver.evaluate_receiver()` is the controlling implementation for one candidate evaluation. Its inputs are a `ReceiverParameters` object, `SimulationConditions`, an `EvaluationFidelity`, a channel path and port map, SKY130 configuration, ngspice configuration, and an optional evaluation cache or eye-capture object. Its output is a `ReceiverEvaluation` containing the stage sequence, success status, measured metrics, violations, warnings, failed stage, runtime, evaluation identity, and provenance.

The evaluator is hierarchical because it increases computational and signal fidelity only after earlier checks succeed. Conceptually, for stages  $s_1, \dots, s_n$ :

$E(\theta, c)$  is the failure record at stage  $s_j$ , if  $s_j$  is the first failed stage; otherwise  $E(\theta, c)$  is the complete metric vector, if all required stages succeed.

Here  $\theta$  is the parameter vector and  $c$  is the operating condition. [ENGINEERING INTERPRETATION] This ordering avoids spending transient, noise, HD3, or PVT runtime on a candidate whose bias or AC response is already invalid.

### 9.2 Fidelity levels

[VERIFIED FROM SOURCE CODE] `EvaluationFidelity` defines SCREENING, TRAINING, CANDIDATE, and FINAL. The receiver uses these levels to select transient bit counts and to enable later stages.

| Fidelity  | Transient request | Channel/receiver role                              | Noise and HD3 role                     |
|-----------|-------------------|----------------------------------------------------|----------------------------------------|
| SCREENING | 32 bits           | Early local circuit screening                      | Not enabled by the receiver stage list |
| TRAINING  | 128 bits          | RL rollout evaluation and channel/receiver metrics | Not enabled by the receiver stage list |
| CANDIDATE | 512 bits          | Candidate qualification metrics                    | Noise and one-amplitude HD3 enabled    |

| Fidelity | Transient request | Channel/receiver role          | Noise and HD3 role                                     |
|----------|-------------------|--------------------------------|--------------------------------------------------------|
| FINAL    | 1024 bits         | Final nominal characterization | Noise and three-amplitude HD3 characterization enabled |

The requested bit count is not necessarily the measured bit count. Warm-up, tail, and integer channel delay are excluded from the valid measurement window. [VERIFIED FROM SOURCE CODE] The historical qualification document reports 960 measured bits for a nominal 1024-bit request. This is a measured artifact-specific result, not a universal count for every condition.

### 9.3 Stage order and stopping behavior

[VERIFIED FROM SOURCE CODE] The runner list is constructed in this order:

1. setup and parameter/condition/model validation;
2. DC operating point;
3. AC response;
4. channel-free CTLE transient;
5. channel diagnostics at TRAINING and above;
6. noise and HD3 at CANDIDATE and above;
7. channel-driven receiver transient and DFE metrics at TRAINING and above.

Each stage returns a `StageResult`. A failed stage stops the sequence and sets `failed_stage`. A simulator failure, parser error, invalid model path, channel preflight failure, or hard metric violation is represented as structured data rather than as an invented metric value.

### 9.4 Setup and validation

`ReceiverParameters.validate()` checks the legacy passive/current variables, DFE tap bounds, and v3 geometry bounds. `SimulationConditions.validate()` checks process-corner type, temperature, supply, common-mode voltage, load, termination, and nonnegative input parasitic capacitance. SKY130 model-library resolution and ngspice identity are checked before the first bench runs.

[VERIFIED FROM SOURCE CODE] A setup failure returns a failed evaluation with a specific failure code such as invalid parameter, missing model library, missing ngspice executable, or channel error. This is important to PPO and to reporting: the simulator failure is observable as a failure stage, not silently treated as an electrical zero.

### 9.5 Nominal filtering and PVT selection

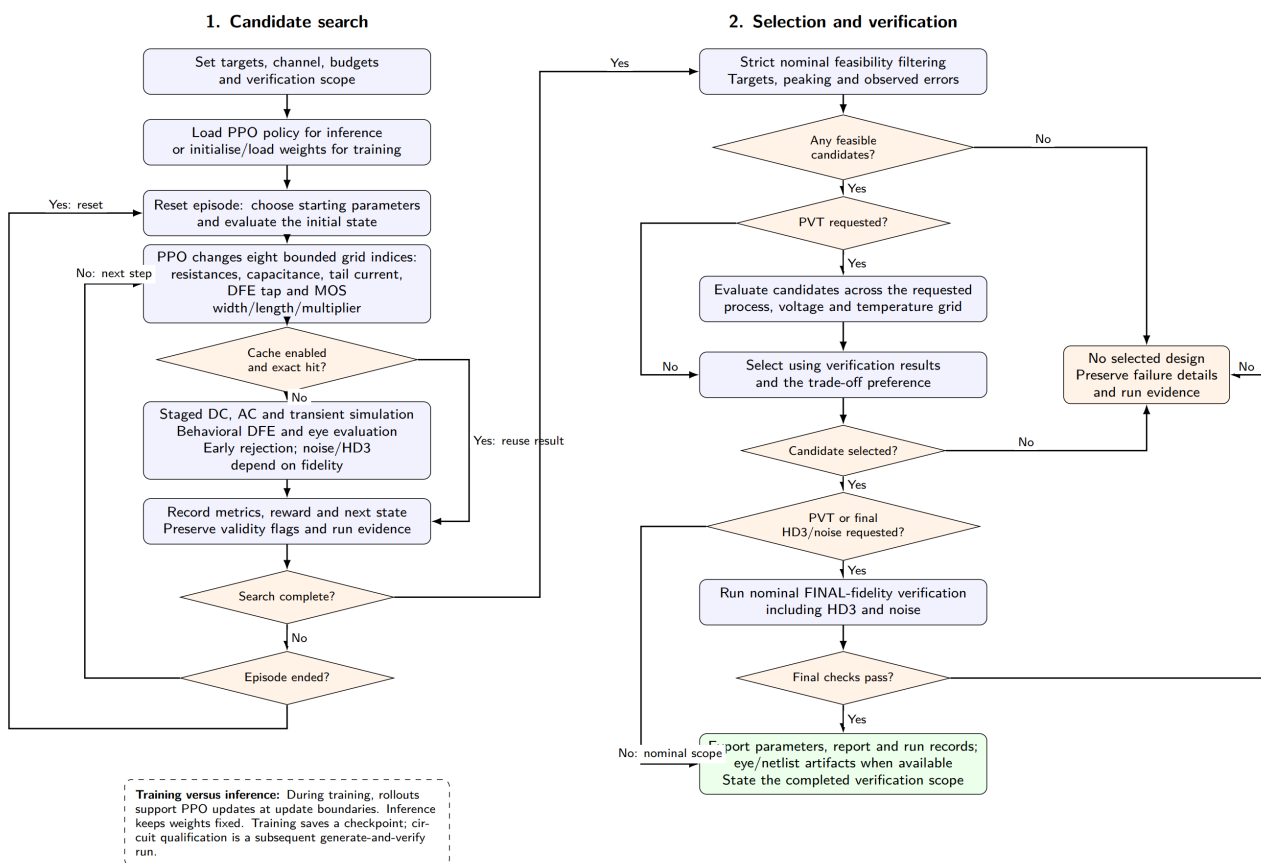
[VERIFIED FROM SOURCE CODE] The pipeline's nominal feasibility filter runs before PVT qualification. Only nominally feasible candidates are sent into the more expensive condition sweep. The PVT selector can rank candidates using nominal validity and available PVT results, but it cannot promote a candidate for conditions that were not evaluated.

[ENGINEERING INTERPRETATION] This separation is a practical compromise: candidate generation needs a cheaper feedback loop, while final acceptance needs broader condition coverage. It also creates a reporting obligation: nominal PASS, candidate-fidelity PASS, and PVT-qualified are different claims and must remain separately labelled.

### 9.6 Stage-result table

| Stage | Main source                                                                       | Inputs                                         | Outputs                              | Failure consequence                           |
|-------|-----------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------|-----------------------------------------------|
| Setup | <code>receiver.py</code> ,<br><code>config.py</code> ,<br><code>ngspice.py</code> | Parameters,<br>conditions,<br>model/tool paths | Validated run context                | No simulation;<br>structured setup<br>failure |
| DC    | <code>ctle_dc.cir</code>                                                          | Bias and model<br>parameters                   | Operating point,<br>power, headroom  | Later stages skipped                          |
| AC    | <code>ctle_ac.cir</code> ,<br><code>waveform.py</code>                            | Differential AC<br>source                      | Gain, peaking, shelf,<br>group delay | Later stages skipped                          |

| Stage              | Main source         | Inputs                    | Outputs                                       | Failure consequence                   |
|--------------------|---------------------|---------------------------|-----------------------------------------------|---------------------------------------|
| CTLE transient     | ctle_transient.cir  | Short NRZ waveform        | Input/output swing                            | Later receiver work skipped           |
| Channel            | channel.py          | S4P and port map          | Loss, delay, passivity, causality diagnostics | Receiver transient skipped            |
| Noise              | ctle_noise.cir      | Small-signal noise setup  | Integrated input-referred noise               | Candidate rejected if threshold fails |
| HD3                | ctle_hd3.cir        | Sinusoidal input          | Third-harmonic ratio                          | Candidate rejected if threshold fails |
| Receiver transient | receiver_transient. | Channel-filtered waveform | Eye, DFE, errors, margin, power               | Candidate rejected if hard gates fail |



**Figure 2.** Optimization, nominal filtering, PVT, final HD3/noise, and reporting flow shown in the supplied presentation. **Evidence:** report\_generation/figures/ppt\_optimization\_flow.png, extracted from the uploaded PPT; [REPORTED IN PPT]. The Python implementation is the controlling source for the exact stage behavior.

## 9.7 Engineering contributions

[VERIFIED FROM SOURCE CODE] The engineering work represented in the repository includes the following concrete components:

- a parameterized SKY130 CTLE subcircuit and reusable SPICE benches;
- Python netlist rendering, ngspice process control, waveform parsing, and structured failure contracts;
- four-port S-parameter parsing, differential-transfer formation, channel passivity/delay/causality screens, and waveform filtering;
- behavioral one-tap DFE sampling, phase selection, eye metrics, margins, and finite-pattern error counting;

- PPO v1, the validity-aware and reward-structured v2 changes, and the versioned eight-parameter v3 contract;
- Random Search and CEM implementations with result consolidation;
- content-addressed evaluation caching, dependency hashing, provenance, checkpointing, resumable training, and event/run-graph records;
- strict PASS/FAIL/NOT CLAIMED reporting, schematic export, dashboard integration, and the natural-language orchestration wrapper;
- unit, integration, synthetic-mechanics, artifact-integrity, and optional real-ngspice tests.

[ENGINEERING INTERPRETATION] These contributions form an evaluation and design-space exploration framework. They do not by themselves constitute a fully physical receiver, a layout database, or an optimizer optimality proof.

## 10. Parameter Space and Design Constraints

### 10.1 Versioned parameter schema

[VERIFIED FROM SOURCE CODE] `simulator.design_schema.py` defines one ordered legacy schema for v1/v2 and one expanded ordered schema for v3. The physical parameter vector is mapped to discrete grids by `rl.parameter_grid.py`; the optimizer therefore operates on grid indices rather than arbitrary floating point values.

| Parameter                   | Meaning                            | Range/Value            | Scaling          | Engineering role                        |
|-----------------------------|------------------------------------|------------------------|------------------|-----------------------------------------|
| <code>r_load_ohm</code>     | Drain-load resistance              | 100 to 10,000 $\Omega$ | Logarithmic      | Gain, bandwidth, output operating point |
| <code>r_deg_ohm</code>      | Source-degeneration resistance     | 10 to 10,000 $\Omega$  | Logarithmic      | Local feedback, linearity, peaking      |
| <code>c_deg_f</code>        | Degeneration capacitance           | 10 fF to 10 pF         | Logarithmic      | Frequency-dependent degeneration bypass |
| <code>i_tail_a</code>       | Tail current                       | 10 $\mu$ A to 1 mA     | Logarithmic      | Bias, transconductance, power           |
| <code>dfe_tap_v</code>      | Behavioral DFE tap                 | -0.4 to 0.4 V          | Linear           | Previous-symbol correction              |
| <code>mos_width_um</code>   | Matched transistor width           | 0.42 to 100 $\mu$ m    | Linear           | Drive, capacitance, partial area        |
| <code>mos_length_um</code>  | Matched transistor length          | 0.15 to 1.0 $\mu$ m    | Linear           | Device behavior and partial area        |
| <code>mos_multiplier</code> | Matched parallel-device multiplier | Integer 1 to 16        | Discrete integer | Drive and partial area                  |

The first five rows are [VERIFIED FROM SOURCE CODE] legacy bounds. The last three are v3 bounds. [VERIFIED FROM SOURCE CODE] Both MOS devices receive the same v3 geometry values; independent unmatched widths or lengths are not exposed by the schema.

### 10.2 Grid representation

The v1/v2/v3 environments use parameter grids with an ordered index for each parameter. For a logarithmically scaled variable  $x$  with bounds  $x_{\min}$  and  $x_{\max}$ , a normalized position can be represented as

$$u = \frac{\ln(x/x_{min})}{\ln(x_{max}/x_{min})}, \quad 0 \leq u \leq 1$$

For a linearly scaled variable,

$$u = \frac{x - x_{min}}{x_{max} - x_{min}}$$

The exact grid-point count and arrays are checkpoint/runtime-contract data. [VERIFIED FROM DOCUMENTATION] The v3 default grid has 21 points for the continuous variables, rounds MOS width and length to 0.001  $\mu\text{m}$ , and uses the integers 1 through 16 for the multiplier. [REQUIRES FURTHER VERIFICATION] Any claim about a candidate outside the checkpoint's stored exact arrays requires checking that checkpoint rather than assuming the default grid.

### 10.3 Action reachability

The default action is not a direct physical assignment. It selects an index delta, which is clipped to the grid:

$$i'_j = \text{clip}(i_j + \Delta_j, 0, N_j - 1)$$

where  $i_j$  is the current index,  $N_j$  is the number of grid points, and  $\Delta_j \in \{-1, 0, +2\}$  by default. [VERIFIED FROM SOURCE CODE] `rl.autockt_action.apply_action()` uses three independent Discrete(3) choices, one per parameter, and clips at the bounds. A boundary action can therefore request movement without changing the applied index.

[ENGINEERING INTERPRETATION] The asymmetric +2 step can reach farther in a short horizon than a symmetric one-step action, but it also makes local movement directionally uneven. The repository includes an optional symmetric ablation using  $(-1, 0, +1)$ ; the default historical contract remains  $(-1, 0, +2)$ .

### 10.4 Constraint types

The parameter bounds are not the same as electrical acceptance criteria. The former define what the optimizer may request; the latter are evaluated by SPICE and receiver analysis. The distinction is:

| Constraint       | Type                           | Example consequence                                 |
|------------------|--------------------------------|-----------------------------------------------------|
| Parameter bounds | Search-domain constraint       | Invalid geometry is rejected before simulation      |
| Grid membership  | Representation constraint      | Candidate must correspond to a stored index         |
| DC/AC gates      | Electrical validity constraint | Candidate stops before receiver transient           |
| Eye/DFE gates    | Receiver constraint            | Candidate fails on errors, margin, height, or width |

| Constraint   | Type                     | Example consequence                           |
|--------------|--------------------------|-----------------------------------------------|
| Area status  | Reporting constraint     | Partial area cannot qualify total-area target |
| PVT coverage | Qualification constraint | Unrun conditions cannot count as passes       |

## 10.5 Parameter-space limitations

[VERIFIED FROM DOCUMENTATION] v3 adds grouped MOS sizing but does not add physical sampler, physical DFE, passive geometry, or layout variables. The search space is therefore a transistor-level CTLE parameter space embedded in a behavioral receiver evaluation, not a complete transceiver layout search.

[NOT AVAILABLE IN THE PROVIDED EVIDENCE] A DRC-feasible geometry grid, extracted parasitic model, passive density model, or layout-aware area constraint is not supplied.

## 11. PPO v1: Baseline Formulation

### 11.1 PPO role in Nebula

Proximal Policy Optimization is used as a sequential candidate generator. At each environment step, a policy maps the current state to independent action choices, the environment applies grid-index deltas, the receiver simulator evaluates the resulting circuit, and the reward is returned to the trainer.

The policy is not a circuit surrogate. [VERIFIED FROM SOURCE CODE] Every training step passes through `ReceiverRLAdapter` and then the real evaluator when the real backend is selected. The policy chooses which designs to try; ngspice determines their measured metrics.

### 11.2 v1 state vector

[VERIFIED FROM SOURCE CODE] `rl.autockt_state.build_state()` constructs a 13-element v1 state from three concatenated components:

$$s_t = [e_t, r_t, i_t]$$

where  $e_t$  is the four-element achieved-versus-target error vector,  $r_t$  is the four-element target-versus-global-reference vector, and  $i_t$  contains the five parameter indices. The normalized error function is

$$\ell(a, g) = \frac{a - g}{a + g}$$

with a small denominator guard in Nebula. Here  $a$  is achieved metric value,  $g$  is the target value, and the result is dimensionless. For the smaller-is-better power specification, the signed error is reversed so that larger positive values consistently indicate better satisfaction.

The direct state-builder documentation retains raw index semantics, while the environment applies a normalized-index adaptation before passing indices into the state construction path. [VERIFIED FROM SOURCE CODE] This distinction is

part of the project's AutoCkt compatibility notes and must not be collapsed into a claim that every direct caller sees the same representation.

### 11.3 v1 action space

[VERIFIED FROM SOURCE CODE] v1 has five independent `Discrete(3)` heads. Choice 0, 1, or 2 selects the default index delta -1, 0, or +2. The complete action is therefore

$$a_t = (a_{t,1}, \dots, a_{t,5}), \quad a_{t,j} \in \{0, 1, 2\}$$

The grid update and clipping equation is the one in Section 10.3. The action mechanics are discrete, but `indices_to_normalized_action()` converts the resulting physical grid values into the continuous [-1,1] encoding expected by the existing receiver adapter.

### 11.4 v1 reward

[VERIFIED FROM SOURCE CODE] `r1.autockt_reward.autockt_reward()` uses the signed relative errors of the four target metrics. It sums only negative unsatisfied contributions and returns a terminal bonus when the total reaches the documented threshold:

`r_t` is defined piecewise:

- `r_t` = the sum of negative excess errors,  $\sum e_j$  for  $j$  with  $e_j < 0$ , when that sum is less than -0.02;
- `r_t` = 10.0, otherwise for a successful evaluation;
- `r_t` = -1.0, if the evaluation fails.

The values -0.02, 10.0, and -1.0 are source-code constants. The last branch is a Nebula adaptation for structured simulator failures, not a claim about the original AutoCkt implementation.

### 11.5 PPO policy and value networks

[VERIFIED FROM SOURCE CODE] `r1.ppo_agent.py` defines a policy network with two hidden layers of 64 units each and tanh activations, followed by one three-logit categorical head per parameter. The value network has a separate two-layer tanh body and a scalar output. The source labels the 64-unit, two-layer shape as AutoCkt-replicated, while the remaining optimizer and network details are explicitly labelled unverified from the original AutoCkt source.

The policy produces categorical distributions

$$\pi_{\phi}(a_t \mid s_t)$$

and the value network estimates

$$V_{\psi}(s_t) \approx \mathbb{E}[\text{return} \mid s_t]$$

Here  $\phi$  and  $\psi$  are policy and value parameters. The exact PPO hyperparameters not established by AutoCkt source are not restated as if they were original-project constants. [NOT AVAILABLE IN THE PROVIDED EVIDENCE] The original AutoCkt source does not pin every value used by the Nebula standalone implementation.

### 11.6 Advantage estimation and clipping

[VERIFIED FROM SOURCE CODE] `rl.trainer.py` collects transitions and `rl.ppo_agent.py` computes generalized advantage estimates with separate terminated and truncated semantics. A conventional form is

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

and

$$\hat{A}_t = \delta_t + \gamma\lambda(1 - d_t)\hat{A}_{t+1}$$

where  $r_t$  is reward,  $\gamma$  is discount,  $\lambda$  is the GAE parameter, and  $d_t$  is one only when the episode truly terminates. For a horizon or budget truncation, the implementation can bootstrap from the reached state's value instead of treating the artificial boundary as a terminal zero.

The PPO policy-ratio objective uses the standard clipped form

$$L^{CLIP}(\phi) = \mathbb{E}_t [\min(r_t(\phi)\hat{A}_t, \text{clip}(r_t(\phi), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)]$$

where  $r_t(\phi) = \pi_\phi(a_t|s_t) / \pi_{\phi_{\text{old}}}(a_t|s_t)$  and  $\epsilon$  is the clipping parameter. [VERIFIED FROM SOURCE CODE] The implementation contains this PPO update structure; exact constants should be taken from the source file for any numerical reproduction.

## 11.7 Episode termination and failure handling

[VERIFIED FROM SOURCE CODE] The environment returns `done=True` on terminal reward success and a separate `truncated=True` flag for horizon or shared evaluation-budget exhaustion. A failed SPICE stage produces a structured failure and reward rather than an uncaught simulator exception in the normal environment path.

[ENGINEERING INTERPRETATION] This distinction matters because PPO's value target should not treat “ran out of allowed steps” as equivalent to “the design is physically terminal.” It also prevents a shared training budget from producing fabricated transitions after the final permitted simulation.

## 11.8 v1 trade-offs and evidence limits

The v1 contract is compact and retains compatibility with the historical five-parameter workflow. Its weaknesses are also explicit: a failed early stage can provide little metric information, unmeasured quantities are not distinguished in the original state representation, and a fixed initial state can limit the observed learning problem.

[VERIFIED FROM DOCUMENTATION] The fair no-warm-start PPO benchmark recorded 0/20 successes at the selected budget, with all episodes failing at DC. This is evidence about that configuration, not proof that PPO cannot solve the design problem under another horizon, target distribution, or state/reward contract.

# 12. PPO v2: State and Reward Improvements

## 12.1 Motivation and compatibility policy

PPO v2 is an additive, opt-in improvement. [VERIFIED FROM DOCUMENTATION] The project preserves v1 behavior when the new environment arguments are omitted. This allows historical v1 checkpoints and result files to remain interpretable while v2 studies the effect of improved simulator information.

The central diagnosis was a reward-information problem: if every sampled episode fails before the four target metrics are genuinely measured, a single failure reward does not tell the policy whether one candidate was closer than another. A second diagnosis was the state-information problem: filling unmeasured metrics with zero can make “not evaluated” resemble “measured zero.”

## 12.2 Corrected reset evaluation

[VERIFIED FROM SOURCE CODE] v2 adds `evaluate_on_reset`. When enabled, the environment evaluates the actual starting circuit before returning the initial state. The state therefore reflects measured starting behavior rather than an artificial all-zero metric dictionary.

This option is intentionally separate from the one-time initial-index randomization feature. The reset contract can preserve a fixed initial grid index for compatibility, while still evaluating that actual point. A reset evaluation consumes simulator budget, so it is not enabled silently for all historical runs.

### 12.3 Validity-aware state schema

[VERIFIED FROM SOURCE CODE] `rl.autockt_state_v2.py` extends the v1 state with:

- one validity flag;
- two early-stage margin dimensions for `ctle_power_w` and `peaking_db`;
- a one-hot failure-stage vocabulary.

The documented dimension is

$$\text{STATE\_DIM}_{v2} = \text{STATE\_DIM}_{v1} + 1 + 2 + 11 = 27$$

The validity flag is not a recomputed guess. It decodes the validity half of the existing adapter observation. The failure-stage one-hot vector includes success, unevaluated reset, setup, DC, AC, CTLE transient, channel, transient, noise, HD3, internal, and the synthetic out-of-region label.

For a metric  $m$ , the validity component is

$q_m = 1$  when  $m$  was genuinely measured, and  $q_m = 0$  when  $m$  was not available.

[ENGINEERING INTERPRETATION] This lets the policy distinguish a measured zero from an unavailable value and distinguish an early DC failure from a late transient near-miss. It does not create a measurement that the simulator never produced.

### 12.4 Structured reward v2

[VERIFIED FROM SOURCE CODE] `rl.autockt_reward_v2.reward_v2()` retains the v1 success contract but adds explicit reward components and available-metric tracking. A failure with no reliable target metrics receives the structural floor `NO_INFORMATION_FLOOR_V2 = -20.0`. The source explains that this is a floor for missing information, not a tuned circuit-performance result.

The v2 reward separates two concepts:

1. `autockt_terminal_success`, which preserves the v1 tolerance-based terminal criterion; and
2. `strict_target_pass`, which checks raw target boundaries independently.

For a valid near-miss, the reward can retain measured constraint margins. For a strict pass, the source adds a bounded overshoot bonus with `OVERSHOOT_BONUS_SCALE = 0.1` and `OVERSHOOT_BONUS_CAP = 2.0`.

[VERIFIED FROM SOURCE CODE] These coefficient values are source constants. They are not presented as experimentally optimized hyperparameters.

### 12.5 Early-stage information

DC and AC stages can produce genuine power and peaking measurements even when the full eye metrics are absent. v2 exposes those measurements as early-stage margins only when they are present in the metrics map. It does not use a fabricated zero for a transient-only metric after a DC failure.

This preserves the failure boundary:

$$\text{available metrics} \subseteq \text{metrics required by later stages}$$

[VERIFIED FROM SOURCE CODE] The module documentation explicitly ties this behavior to the receiver's existing stage results and validity flags.

### 12.6 v1, v2, and v3 comparison

| Feature                | PPO v1                                              | PPO v2                               | PPO v3                                               | Evidence                                    |
|------------------------|-----------------------------------------------------|--------------------------------------|------------------------------------------------------|---------------------------------------------|
| Optimized parameters   | 5                                                   | 5                                    | 8                                                    | [VERIFIED FROM SOURCE CODE]                 |
| State dimension        | 13                                                  | 27                                   | 52                                                   | [VERIFIED FROM SOURCE CODE] / documentation |
| Reset measurement      | Optional historical behavior; default no evaluation | Opt-in actual reset evaluation       | Versioned v3 contract                                | [VERIFIED FROM SOURCE CODE]                 |
| Validity mask          | Not represented in v1 state                         | Explicit validity information        | Per-metric validity plus v3 features                 | [VERIFIED FROM SOURCE CODE]                 |
| Failure-stage encoding | Not represented in v1 state                         | One-hot stage vocabulary             | Retained and extended in v3 state                    | [VERIFIED FROM SOURCE CODE]                 |
| Reward                 | Flat failure reward and terminal bonus              | Structured valid/failure reward      | Strict receiver reward plus partial-area regularizer | [VERIFIED FROM SOURCE CODE]                 |
| Checkpoint contract    | Historical bare state dictionary                    | Compatibility-preserving legacy form | Metadata-bearing versioned export                    | [VERIFIED FROM SOURCE CODE]                 |

The v3 column is included for continuity; its full geometry and runtime contract are developed in Section 13. [REQUIRES FURTHER VERIFICATION] The state dimensions are implementation contract values and should be checked against the exact checkpoint metadata when loading an individual artifact.

## 12.7 Evidence and limitations

The v2 design directly addresses observed information loss and simulator failure handling. It does not demonstrate, by itself, that real-circuit PPO outperforms Random Search or CEM. A controlled ablation file exists and the test suite verifies its mechanics, but broad statistical conclusions require the recorded seed count, budget, target pool, and backend to be reported with each result.

[NOT AVAILABLE IN THE PROVIDED EVIDENCE] A multi-seed, matched-budget real- SPICE study proving a statistically reliable v2 improvement over v1 is not available as a completed general claim.

## 13. PPO v3: Expanded Sizing and Runtime Contract

### 13.1 Motivation for v3

[VERIFIED FROM DOCUMENTATION] PPO v3 expands the optimization space from the five legacy variables to eight variables by adding a matched MOS width, matched MOS length, and integer MOS multiplier. This is a meaningful model change, not a relabeling of the v1/v2 checkpoint. The v3 contract therefore uses a new state/action/checkpoint schema and rejects incompatible legacy policies.

[ENGINEERING INTERPRETATION] The expansion addresses a physical limitation of the earlier search: passive degeneration, load, bias, and behavioral DFE settings could be adjusted, but transistor geometry remained fixed. It also increases the search difficulty because each geometry changes both electrical behavior and partial area.

### 13.2 v3 state construction

[VERIFIED FROM SOURCE CODE] `rl/autockt_v3.py::build_state_v3()` combines the following state groups:

- target error and target-reference vectors;
- eight normalized parameter indices;
- normalized values for the v3 metric vocabulary;
- per-metric availability flags;

- one-hot failure-stage identity;
- channel loss features at 1.25, 2.5, and 5 GHz and fitted bulk delay;
- a normalized partial MOS channel-area feature;
- an explicit false total-area-valid flag.

The documented v3 dimension is 52. The exact composition is checkpoint contract data and is not silently inferred when loading a policy. The channel features are calculated from the selected channel path and port map, so a policy's state includes information about the channel under which it is being evaluated.

### 13.3 v3 reward and termination

[VERIFIED FROM SOURCE CODE] `reward_v3()` requires successful evaluation, valid target metrics, all target specifications satisfied, CTLE peaking in the 3--12 dB band, and zero measured DFE decision errors. It adds a small partial-MOS-area regularizer only when that metric is available. The area term is not a feasibility gate.

The strict success condition can be represented as

$$S_{v3} = S_{sim} \wedge S_{target} \wedge S_{peak} \wedge (N_{err} = 0)$$

where  $S_{sim}$  means required simulation stages completed,  $S_{target}$  is the strict target assessment,  $S_{peak}$  is the peaking-band condition, and  $N_{err}$  is the measured finite-pattern error count.

[ENGINEERING INTERPRETATION] Separating reward shaping from final area qualification is important. A smaller partial MOS channel area may influence ranking during learning, but the system must not convert it into a complete 0.05 mm<sup>2</sup> receiver-area claim while passive and physical DFE areas are absent.

### 13.4 Checkpoint compatibility

[VERIFIED FROM SOURCE CODE] v3 inference exports contain metadata describing the state schema, reward schema, ordered parameters, grids, action deltas, channel contract, and toolchain context. The runtime contract checks these fields before inference. A bare v1/v2 state dictionary is rejected under v3 rather than being interpreted with the wrong input dimension or parameter meaning.

| Checkpoint type     | Metadata requirement                                      | Loading status                                   |
|---------------------|-----------------------------------------------------------|--------------------------------------------------|
| Legacy v1/v2 policy | Historical bare state dictionary                          | Valid only under compatible legacy loading path  |
| v3 inference policy | Metadata-bearing policy export                            | Accepted by v3 contract when fields match        |
| v3 full checkpoint  | Policy, value, optimizer, RNG, counters, grids, toolchain | Resumable only under compatible runtime contract |

[VERIFIED FROM RAW SIMULATION DATA] `results/ppo_v3_tt_1000_policy.pt` is a genuine metadata-bearing v3 policy associated with the real-backend summary `results/v3_tt_1000_summary.json`. [VERIFIED FROM LOGS] Direct loading and candidate generation succeeded under the v3 compatibility check.

### 13.5 Resumption and reproducibility

[VERIFIED FROM DOCUMENTATION] Full v3 checkpoints preserve policy and value weights, optimizer state, Python/NumPy/Torch random state, target/reset RNG, update and evaluation counters, training settings, exact grids, channel checksum/map, and toolchain identity. The in-memory evaluation cache is not serialized, so resuming can repeat simulator work.

The distinction between `--updates` and `--max-evaluations` is material: updates represent additional learning iterations, while the total evaluation budget includes work already performed before resume. [ENGINEERING INTERPRETATION] This prevents a resumed run from accidentally receiving a new unreported simulation budget merely because its update count is reset.

### 13.6 v3 evidence and limits

The real v3 artifact records 100 updates and 692 evaluations with seed 101. That demonstrates an executed real-backend training workflow. It does not, without a matched multi-seed study, demonstrate convergence, superiority, or generalization.

[NOT AVAILABLE IN THE PROVIDED EVIDENCE] The literal original shell command used to produce the historical v3 checkpoint is not recorded. A reconstruction from embedded metadata is possible, but it must remain labelled as a reconstruction rather than an original command transcript.

## 14. Random Search and CEM Baselines

### 14.1 Purpose of baseline comparison

Random Search and CEM provide non-PPO references for the same design space. They are useful only when parameter order, bounds, evaluation fidelity, target, budget, initialization, and success definition are comparable. [VERIFIED FROM SOURCE CODE] The repository retains separate implementations and result schemas for both baselines.

### 14.2 Random Search

[VERIFIED FROM SOURCE CODE] `experiments/receiver_search.py` samples each normalized parameter independently from a uniform interval:

$$u_j \sim U(-1, 1)$$

where  $u_j$  is the normalized coordinate for parameter  $j$ . The coordinate is then decoded using the parameter's linear or logarithmic scaling. The method does not learn from earlier candidates.

[ENGINEERING INTERPRETATION] Random Search has broad coverage and simple reproducibility, but it spends evaluations without using measured gradients, failure stage, or local response. In a sparse feasible region, success probability depends strongly on the initial distribution and budget.

### 14.3 Cross-Entropy Method

[VERIFIED FROM SOURCE CODE] `experiments/train_cem.py` samples a population from a factorized Gaussian in normalized coordinates. The documented defaults include initial mean zero, initial standard deviation 0.15, population five, and elite count two for the relevant CEM configuration. After scoring a population, the elite mean and standard deviation are updated, with a standard-deviation floor retained by the implementation. A single-elite fallback shrinks the spread by the documented factor.

For normalized coordinate vector  $u \in \mathbb{R}^d$ ,

$$u \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$$

and for elite set  $E$ ,

$$\mu' = \frac{1}{|E|} \sum_{u \in E} u, \quad \sigma'_j = \max \left( \sigma_{\min}, \sqrt{\frac{1}{|E|} \sum_{u \in E} (u_j - \mu'_j)^2} \right)$$

The equations describe the implemented family of update; exact population, elite, and fallback values are controlled by the source and run configuration.

## 14.4 Comparison evidence

| Optimizer         | Search method                          | Budget                               | Best result  | Runtime                               | Evidence                                                         |
|-------------------|----------------------------------------|--------------------------------------|--------------|---------------------------------------|------------------------------------------------------------------|
| Random Search     | Independent uniform normalized samples | 20 candidates in fair trial          | 1/20 success | Not available in the comparison table | [VERIFIED FROM LOGS]                                             |
| CEM               | Gaussian population and elite update   | 20 candidates in fair trial          | 0/20 success | Not available in the comparison table | [VERIFIED FROM LOGS]                                             |
| PPO               | Sequential discrete policy actions     | 20 candidates in fair trial          | 0/20 success | Not available in the comparison table | [VERIFIED FROM LOGS]                                             |
| CEM, warm-started | CEM near verified design               | 13/30 planned before incomplete stop | 8/13 success | Not available in the comparison table | [VERIFIED FROM LOGS]<br>Not comparable to uniform Random Search. |

[VERIFIED FROM DOCUMENTATION] The fair head-to-head is one seed and carries a statistical-power caveat. The warm-started CEM result is not a fair advantage claim because initialization and completion differ.

## 14.5 Baseline limitations

[ENGINEERING INTERPRETATION] A 0/20 result can reflect reachability, short horizon, early DC failure, target difficulty, or insufficient budget. It is not enough to conclude an algorithm is intrinsically incapable. A stronger comparison requires multiple independent seeds, identical ordered schemas, matched evaluation budgets, raw metrics for every candidate, wall-clock timing, and predeclared success criteria.

[NOT AVAILABLE IN THE PROVIDED EVIDENCE] A statistically powered, multi-seed, real-SPICE comparison establishing a reliable ordering among PPO, Random Search, and CEM is not present.

# 15. Caching, Provenance, Checkpoints, and Run Graphs

## 15.1 Content-addressed evaluation cache

[VERIFIED FROM SOURCE CODE] `simulator.cache.EvaluationCache` stores scalar evaluation results under content-derived identifiers. The receiver identity includes parameters, operating conditions, fidelity, implementation hashes, bench hashes, channel identity, model-library dependencies, simulator identity, and channel port map where relevant.

Conceptually,

$$ID = H(\theta, c, F, I_{impl}, I_{model}, I_{bench}, I_{channel}, I_{tool})$$

where  $H$  is the stable fingerprint function and the  $I$  terms are content identities. A cache hit is safe only when all those inputs match.

[VERIFIED FROM SOURCE CODE] Cache writes use an atomic lock-file protocol. Malformed or obsolete cache entries are treated as misses rather than fatal results. Eye-capture evaluations bypass the scalar cache because they require additional bounded waveform data.

## 15.2 Provenance

`simulator.provenance.py` records Git identity, SHA-256 file hashes, model dependency closure, ngspice executable/version, solver selection, and schema versions. [VERIFIED FROM SOURCE CODE] The evaluator hashes all bench files and the implementation modules that control receiver metrics. The purpose is to prevent a result produced by one code/model/toolchain state from being mistaken for a result produced by another.

### 15.3 Checkpoint contents

Checkpointing operates at two levels:

- inference-only policy exports for candidate generation;
- full training checkpoints for resume.

[VERIFIED FROM SOURCE CODE] Full checkpoints preserve optimizer and RNG state as well as model weights and runtime contract information. v3 inference requires metadata, whereas historical bare policy files remain part of the legacy compatibility surface.

### 15.4 Run graphs and structured events

[VERIFIED FROM DOCUMENTATION] The UI and training path write per-step events and execution graphs containing reset calls, actions, parameter changes, stages, cache origins, failures, skipped stages, selection, verification, and artifacts. JSON retains the complete graph; the visual overview is limited to 180 nodes.

[ENGINEERING INTERPRETATION] A run graph complements scalar metrics by showing why a candidate reached or failed a stage. It is especially useful for distinguishing a policy decision, cache reuse, simulator failure, and qualification rejection.

### 15.5 Artifact integrity

[VERIFIED FROM SOURCE CODE] The Phase 2 artifact package includes a SHA-256 ledger and a test that compares every archived member with the recorded digest. The repository also preserves crash logs, JSONL experiment records, plots, schematics, and checkpoint files rather than replacing historical failures with later successful runs.

## 16. Nominal Real-SPICE Results

### 16.1 Fresh measured result

[VERIFIED FROM RAW SIMULATION DATA] The fresh evidence file `results/llm_wrapper_runs/smoke_evidence_demo.json` records the following nominal result for the selected design and its stated TT/1.8 V/27 °C context.

| Metric                  | Value  | Unit | Source           | Verification Status                 | Interpretation                                       |
|-------------------------|--------|------|------------------|-------------------------------------|------------------------------------------------------|
| Eye width               | 0.73   | UI   | smoke_evidence_c | [VERIFIED FROM RAW SIMULATION DATA] | Exceeds the implemented candidate gate for that run. |
| Locked-phase eye height | 1.55   | V    | Same             | [VERIFIED FROM RAW SIMULATION DATA] | Vertical opening at selected phase.                  |
| Decision margin         | 0.7294 | V    | Same             | [VERIFIED FROM RAW SIMULATION DATA] | Minimum signed sampled margin.                       |

| Metric                   | Value                  | Unit            | Source | Verification Status                 | Interpretation                                   |
|--------------------------|------------------------|-----------------|--------|-------------------------------------|--------------------------------------------------|
| CTLE power               | 1.206                  | mW              | Same   | [VERIFIED FROM RAW SIMULATION DATA] | CTLE-level power, not total receiver power.      |
| AC peaking               | 3.599                  | dB              | Same   | [VERIFIED FROM RAW SIMULATION DATA] | AC metric in the implemented peaking definition. |
| HD3                      | -75.62                 | dB              | Same   | [VERIFIED FROM RAW SIMULATION DATA] | 100 mVpp nominal HD3 result.                     |
| Input-referred noise     | 0.372                  | mVrms           | Same   | [VERIFIED FROM RAW SIMULATION DATA] | Integrated noise over the implemented band.      |
| DFE errors               | 0                      | count           | Same   | [VERIFIED FROM RAW SIMULATION DATA] | Zero in the finite measured pattern.             |
| Partial MOS channel area | $3.113 \times 10^{-6}$ | mm <sup>2</sup> | Same   | [VERIFIED FROM RAW SIMULATION DATA] | Explicitly partial; not total-area evidence.     |

## 16.2 Result interpretation

[ENGINEERING INTERPRETATION] The nominal result demonstrates that one metadata-compatible v3 policy can generate a candidate that reaches the receiver-level gates and can be refined with noise and HD3 measurements. It does not show that every policy rollout succeeds, that the design is best, or that the nominal point remains valid over untested corners.

[VERIFIED FROM LOGS] The fresh nominal-only pipeline runtime was 330.67 s and the smoke-evidence run with additional HD3/noise and PVT-smoke work was 1056.52 s. These times are run-specific and should not be generalized to all machines or all candidates.

## 16.3 Historical specification artifact

[VERIFIED FROM RAW SIMULATION DATA] `results/design_a_final_specification.json` contains PASS rows for measurable eye, power, peaking, HD3, and noise metrics, while the transistor channel-area row is marked NOT CLAIMED for the complete area requirement. This artifact is authoritative for its design and run context; it is not evidence that total receiver area was measured.

## 16.4 Nominal result boundaries

The result is conditioned on the selected model library, channel, source and termination settings, stimulus, fidelity, target, and checkpoint. [REQUIRES FURTHER VERIFICATION] A result copied without those fields is not fully reproducible. No silicon or post-layout measurement is associated with the nominal JSON file.

## 16.5 User-supplied wrapper execution screenshot

[REPORTED IN PPT] A user-supplied screenshot titled by the user as proof of the working wrapper shows a completed real-backend v3 execution. The visible console record reports exit code 0, runtime 412.90 s, checkpoint

results/ppo\_v3\_tt\_1000\_policy.pt, and the following selected parameters and measured rows:

| Metric               | Visible value | Unit    | Visible status     | Evidence interpretation                                                               |
|----------------------|---------------|---------|--------------------|---------------------------------------------------------------------------------------|
| rload_ohm            | 2457.14       | ohm     | Selected parameter | [REPORTED IN PPT] Screenshot value; repository artifact remains the primary source.   |
| rdeg_ohm             | 485.714       | ohm     | Selected parameter | [REPORTED IN PPT]                                                                     |
| cdeg_f               | 4.85714e-13   | F       | Selected parameter | [REPORTED IN PPT]                                                                     |
| itail_a              | 0.00067       | A       | Selected parameter | [REPORTED IN PPT]                                                                     |
| dfe_tap_v            | 0.0571429     | V       | Selected parameter | [REPORTED IN PPT]                                                                     |
| mos_width_um         | 10.378        | um      | Selected parameter | [REPORTED IN PPT]                                                                     |
| mos_length_um        | 0.15          | um      | Selected parameter | [REPORTED IN PPT]                                                                     |
| mos_multiplier       | 1             | integer | Selected parameter | [REPORTED IN PPT]                                                                     |
| Eye width            | 0.76          | UI      | PASS               | Visible wrapper result; differs from other run artifacts and is not merged with them. |
| Eye height           | 1.545         | V       | PASS               | Visible wrapper result; DFE/receiver metric as labelled by the wrapper.               |
| Margin               | 0.7281        | V       | PASS               | Visible wrapper result.                                                               |
| Power                | 0.001206      | W       | PASS               | Visible CTLE power row, not total receiver power.                                     |
| Peaking              | 3.599         | dB      | PASS               | Visible AC metric.                                                                    |
| HD3                  | n/a           | —       | NOT CLAIMED        | Screenshot explicitly says not executed, not a pass.                                  |
| Input-referred noise | n/a           | —       | NOT CLAIMED        | Screenshot explicitly says not executed, not a pass.                                  |

| Metric | Visible value | Unit | Visible status | Evidence interpretation                              |
|--------|---------------|------|----------------|------------------------------------------------------|
| PVT    | n/a           | —    | NOT CLAIMED    | Screenshot explicitly says not executed, not a pass. |

[Figure unavailable: source image not found in the available project evidence]

**Figure 3.** User-supplied wrapper console output showing a successful real- backend v3 pipeline invocation, selected circuit parameters, measured nominal PASS rows, and explicit NOT CLAIMED rows for unrequested HD3, noise, and PVT. **Evidence:** user-supplied attachment “Pasted text #1”; the binary image was not available as a local project asset, so no image is embedded above. This screenshot’s text transcript proves a completed wrapper invocation and its displayed verdicts; it does not independently prove the underlying raw waveform or model-library provenance.

[ENGINEERING INTERPRETATION] The screenshot is useful interface evidence because it shows the wrapper returning a structured result rather than merely printing a generic success message. Its values must remain a separate run snapshot unless matched to a raw JSON artifact by checkpoint, target, channel, conditions, and evaluation identity.

The screenshot shows a dashboard with two main sections. The top section, titled 'FINAL CIRCUIT PARAMETERS', contains eight cards with the following data:

| Parameter      | Value                 |
|----------------|-----------------------|
| rload_ohm      | 2457.1428571428573    |
| rdeg_ohm       | 485.7142857142857     |
| cdeg_f         | 4.857142857142856e-13 |
| tail_a         | 0.00067               |
| dfc_deg_v      | 0.05714285714285683   |
| mos_width_um   | 10.378                |
| mos_length_um  | 0.15                  |
| mos_multiplier | 1                     |

The bottom section, titled 'MEASURED SPECIFICATION', contains a table with the following data:

| Metric                         | Measured  | Requirement                                                                                                                                                                                       | Verdict     |
|--------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Eye width (UI)                 | 0.9       | > 0.4 UI                                                                                                                                                                                          | PASS        |
| Eye height (V)                 | 1.392     | > 0.1 V (this repo's own EXISTING_THRESHOLD value -- the official slide's exact mV figure was not independently re-verified against this number in this pass; see docs/autockt-mapping.md sec 21) | PASS        |
| Margin (V)                     | 0.6594    | > 0 V                                                                                                                                                                                             | PASS        |
| Power (W)                      | 0.001206  | 0 < power < 0.015 W (15 mW)                                                                                                                                                                       | PASS        |
| Peaking (dB)                   | 3.599     | 3-12 dB, -1.25-2.5 GHz                                                                                                                                                                            | PASS        |
| HD3 (dB)                       | -75.62    | < -30 dB                                                                                                                                                                                          | PASS        |
| Input-referred noise (Vrms)    | 0.000372  | < 0.0015 Vrms (1.5 mV)                                                                                                                                                                            | PASS        |
| Transistor channel area (mm^2) | 3.113e-06 | < 0.05 mm^2 -- PARTIAL measurement only (channel area, not total circuit area; resistor/capacitor/layout-overhead area are not modeled anywhere in this project -- see analysis/area_estimate.py) | NOT CLAIMED |
| PVT (pass/total)               | 27/27     | TT/SS/FF x 1.71/1.80/1.89 V x 0.27/75/125 C (agreed 36-condition grid)                                                                                                                            | NOT CLAIMED |

**Figure 4.** Dashboard final-parameter cards and measured-specification table from a captured Nebula run — the one supplied image that shows genuine application interface chrome (labelled parameter fields and a verdict table), as distinct from the regenerated analysis charts in Section 19. **Evidence:** report\_generation/figures/ppt\_final\_parameters\_specification.png, extracted from the uploaded PPT; [REPORTED IN PPT]. The visible PASS/NOT CLAIMED rows are interface output for that captured run and are not merged with the fresh JSON result table above.

### Final Selected Candidate

The table below records the candidate visible in the supplied working-wrapper execution screenshot. It is kept separate from the fresh JSON result because the screenshot reports a different eye width and runtime and does not expose all provenance fields.

| Field  | Value       | Unit/status | Evidence                                              |
|--------|-------------|-------------|-------------------------------------------------------|
| R_LOAD | 2457.14     | ohm         | [REPORTED IN PPT]<br>User-supplied wrapper screenshot |
| R_DEG  | 485.714     | ohm         | [REPORTED IN PPT]                                     |
| C_DEG  | 4.85714e-13 | F           | [REPORTED IN PPT]                                     |
| I_TAIL | 0.00067     | A           | [REPORTED IN PPT]                                     |
| V_DFE  | 0.0571429   | V           | [REPORTED IN PPT]                                     |
| W_MOS  | 10.378      | um          | [REPORTED IN PPT]                                     |
| L_MOS  | 0.15        | um          | [REPORTED IN PPT]                                     |
| M_MOS  | 1           | integer     | [REPORTED IN PPT]                                     |

| Field                                                | Value                         | Unit/status     | Evidence                                 |
|------------------------------------------------------|-------------------------------|-----------------|------------------------------------------|
| Optimizer/checkpoint results/po_v3_tt_1000_policy.pt | PPO v3 /                      | —               | [REPORTED IN PPT]                        |
| Random seed                                          | Unavailable in the screenshot | —               | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Channel                                              | Unavailable in the screenshot | —               | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Model library                                        | Unavailable in the screenshot | —               | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Supply                                               | Unavailable in the screenshot | —               | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Temperature                                          | Unavailable in the screenshot | —               | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Simulation fidelity                                  | Unavailable in the screenshot | —               | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Wrapper runtime                                      | 412.90                        | s               | [REPORTED IN PPT]                        |
| Eye height                                           | 1.545                         | V               | PASS shown; DFE/receiver metric          |
| Eye width                                            | 0.76                          | UI              | PASS shown; DFE/receiver metric          |
| HD3                                                  | Unavailable; NOT CLAIMED      | —               | [REPORTED IN PPT] Not executed           |
| Input-referred noise                                 | Unavailable; NOT CLAIMED      | —               | [REPORTED IN PPT] Not executed           |
| Power                                                | 0.001206                      | W               | PASS shown; CTLE-level power             |
| Partial MOS channel area                             | 3.113e-06                     | mm <sup>2</sup> | Explicitly partial area                  |
| PVT status                                           | Unavailable; NOT CLAIMED      | —               | [REPORTED IN PPT] Not executed           |

[ENGINEERING INTERPRETATION] This is a selected candidate snapshot, not a universal final-design claim. Complete area, total receiver power, SF/FS PVT, HD3, and noise cannot be inferred from the screenshot's nominal PASS rows.

## 17. PVT Qualification Results

### 17.1 Condition sets

[VERIFIED FROM SOURCE CODE] The full grid is

**5 corners × 3 supplies × 4 temperatures = 60 conditions**

with corners TT, SS, FF, SF, FS; supplies 1.71, 1.80, and 1.89 V; and temperatures 0, 27, 75, and 125 °C. The agreed qualification grid is

# 3 corners × 3 supplies × 4 temperatures = 36 conditions

using TT, SS, and FF. The minimal historical grid contains 27 TT/SS/FF conditions with the documented temperature subset.

## 17.2 Historical finalist result

[VERIFIED FROM DOCUMENTATION] The dated v3 qualification artifact reports:

| Design       | Nominal status                            | Corner result | Status                            | Verification                                   |
|--------------|-------------------------------------------|---------------|-----------------------------------|------------------------------------------------|
| pipeline_ep0 | PASS at nominal 512 and 1024-bit contexts | 36/36         | Qualified for the documented grid | [VERIFIED FROM RAW SIMULATION DATA] Historical |
| pipeline_ep1 | PASS at nominal contexts                  | 18/36         | Rejected by PVT                   | [VERIFIED FROM RAW SIMULATION DATA] Historical |

The corner ranges recorded for pipeline\_ep0 include HD3 from -79.8428 to -69.6987 dB, input-referred noise from 0.000328262 to 0.00048729 Vrms, eye height from 1.27665 to 1.65185 V, eye width from 0.71 to 0.78 UI, CTLE power from 0.0011457 to 0.0012663 W, and zero observed errors across the reported conditions. [VERIFIED FROM RAW SIMULATION DATA] These values are historical candidate-fidelity results, not a fresh final-fidelity rerun.

## 17.3 Qualification semantics

A PVT pass is a conjunction over the evaluated condition set:

$$Q(D) = \bigwedge_{c \in C} S(D, c)$$

where D is a design, C is the required condition set, and S(D,c) is the strict stage and metric success for condition c. If a condition is not run, it cannot contribute a PASS. If a run is interrupted or times out, the condition remains incomplete or failed according to the result contract.

[VERIFIED FROM DOCUMENTATION] The historical artifact records 36/36 on the TT/SS/FF grid. [NOT AVAILABLE IN THE PROVIDED EVIDENCE] A completed full 60-condition qualification is not present.

## 17.4 Fresh final-fidelity status

[VERIFIED FROM LOGS] A recorded final-fidelity run completed and passed three of 36 conditions before it was stopped. The measured pace was approximately eight minutes per condition. This is evidence of runtime cost and partial progress, not a complete final-fidelity qualification.

## 17.5 Why both 27 and 36 appear in the evidence

The 27-point and 36-point results are different condition sets, not duplicate claims. [VERIFIED FROM RAW SIMULATION DATA] The preserved design\_a\_pvt\_minimal27.jsonl and \_rerun.jsonl files each contain 27 conditions covering TT/SS/FF, 1.71/1.80/1.89 V, and temperatures 0, 27, and 125 °C. The 75 °C row is absent from that minimal set.

[VERIFIED FROM RAW SIMULATION DATA] The v3 qualification artifact contains 36 required conditions covering the same three process corners and supply values, but temperatures 0, 27, 75, and 125 °C. The historical pipeline\_ep0 result is therefore 36/36 on the four-temperature TT/SS/FF grid. The 27-condition heatmaps describe the separate

minimal historical grid and should not be read as 36-point qualification or as five-corner 60-point qualification.

[VERIFIED FROM SOURCE CODE] The full defined grid adds SF and FS and contains 60 conditions. No artifact in the provided evidence establishes 60/60.

## 18. PPO Learning and Optimizer Comparison

### 18.1 Training evidence

[VERIFIED FROM RAW SIMULATION DATA] The v3 summary records a completed real- backend training run with seed 101, 100 updates, and 692 evaluations. Its best-reward and candidate information are preserved in the summary JSON.

[ENGINEERING INTERPRETATION] This proves that the training loop can execute through real evaluation and save a usable v3 policy. It does not establish that the policy converged globally or that learning caused the eventual success rather than the initialization, target, action reachability, or selection process.

### 18.2 Fair comparison

| Optimizer     | Search method               | Budget | Best result | Runtime                            | Evidence             |
|---------------|-----------------------------|--------|-------------|------------------------------------|----------------------|
| PPO           | Sequential discrete policy  | 20     | 0/20        | Not recorded in comparison summary | [VERIFIED FROM LOGS] |
| Random Search | Uniform normalized sampling | 20     | 1/20        | Not recorded in comparison summary | [VERIFIED FROM LOGS] |
| CEM           | Gaussian elite update       | 20     | 0/20        | Not recorded in comparison summary | [VERIFIED FROM LOGS] |

[VERIFIED FROM DOCUMENTATION] The study used one seed and carries an explicit low-statistical-power caveat. The correct conclusion is that PPO superiority was not demonstrated under this setup, not that PPO is universally inferior.

### 18.3 Unseen-target and frozen-policy evidence

[VERIFIED FROM RAW SIMULATION DATA] The controlled unseen-target artifact records regression described as one win, six ties, and three losses for the trained policy comparison. The project therefore does not claim reliable generalization to unseen targets.

[VERIFIED FROM DOCUMENTATION] The v3 midpoint comparison evaluates frozen policies and excludes pretraining cost. It is evidence about a specific frozen-policy experiment, not a multi-seed learning comparison.

### 18.4 Learning conclusion

The defensible claim is narrow: Nebula implements PPO training, preserves versioned checkpoints, and has real-backend training and inference evidence. The evidence does not support a stronger claim that PPO is the most efficient optimizer, generalizes reliably, or removes the need for baseline search.

## What the Optimizer Experiments Establish

[VERIFIED FROM SOURCE CODE] PPO, Random Search, and CEM are integrated into the same broader receiver-evaluation workflow. [VERIFIED FROM LOGS] The fair head-to-head record is PPO 0/20, Random Search 1/20, and CEM 0/20 under one seed and a matched small budget. A separate warm-started CEM result is not comparable to the unbiased trial.

These experiments establish implementation and feasibility evidence: each optimizer can consume the project parameter schema and submit candidates to the evaluation pipeline. They do not establish PPO superiority, near- optimality, or global optimality. The comparison is limited by seed count, evaluation budget, incomplete raw-data availability for some historical runs, initialization asymmetry in non-controlled trials, and statistical power.

[ENGINEERING INTERPRETATION] The correct use of the comparison is to test whether the optimizer integrations work and whether feasible designs can be found under stated budgets. A conclusive optimizer ranking would require matched multi-seed real-SPICE experiments, identical initialization policy, raw metrics for every request, wall-clock measurements, and a predeclared statistical analysis.

## 19. Eye Diagrams, AC Response, HD3, and Noise

### 19.1 Eye artifacts

[VERIFIED FROM RAW SIMULATION DATA] The repository contains `results/v3_tt_1000_eye.png`, SVG, and bounded eye-source JSON artifacts. The source data distinguish a continuous transistor-level CTLE output eye from behavioral DFE corrected samples. The DFE panel must not be captioned as a transistor-level waveform.

The eye artifact is generated after selection when eye capture is requested; training does not capture and render eyes for every PPO step. Missing waveform data are reported as not produced rather than fabricated.

### 19.2 AC response

[VERIFIED FROM SOURCE CODE] AC data are produced by the logarithmic 1 MHz to 20 GHz bench and reduced to scalar gains, peaking, response class, shelf metrics, and group-delay metrics. The historical tuning artifact reports three target frequencies, with one of three measured points within  $\pm 5\%$ .

| Target    | Measured              | Within $\pm 5\%$ | Evidence                            |
|-----------|-----------------------|------------------|-------------------------------------|
| 1.25 GHz  | 1.320527183368927 GHz | No               | [VERIFIED FROM RAW SIMULATION DATA] |
| 1.875 GHz | 1.908910958457706 GHz | Yes              | [VERIFIED FROM RAW SIMULATION DATA] |
| 2.5 GHz   | 2.295117021187582 GHz | No               | [VERIFIED FROM RAW SIMULATION DATA] |

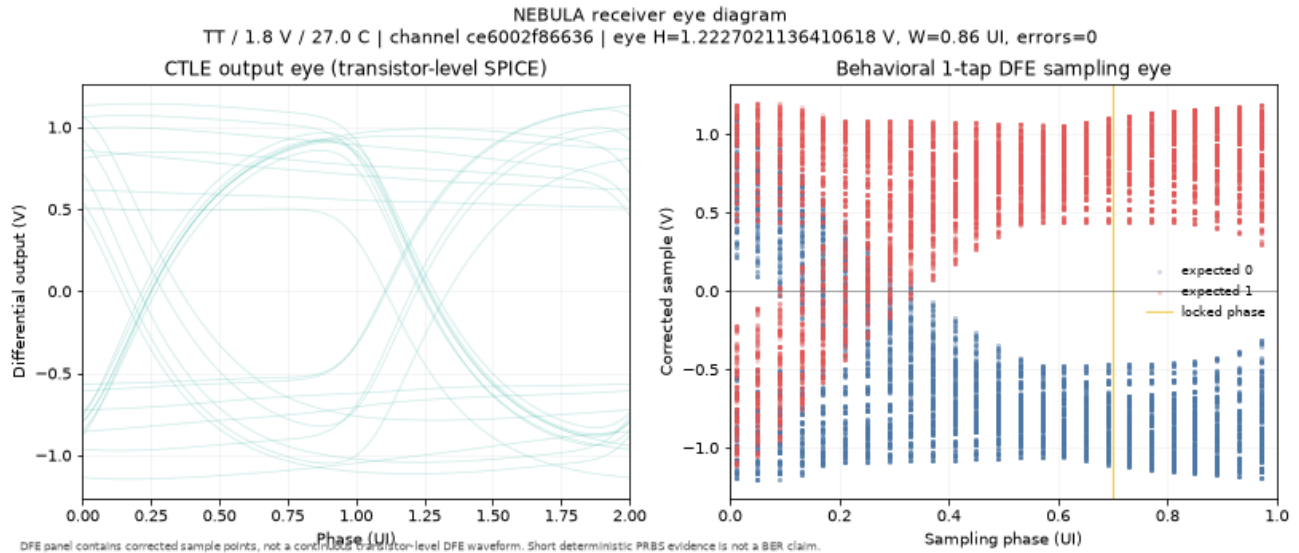
This is sampled nominal tuning evidence. [REQUIRES FURTHER VERIFICATION] Continuous tuning compliance and receiver/noise/HD3 compliance at every tuning setting are not demonstrated.

### 19.3 HD3 and noise evidence

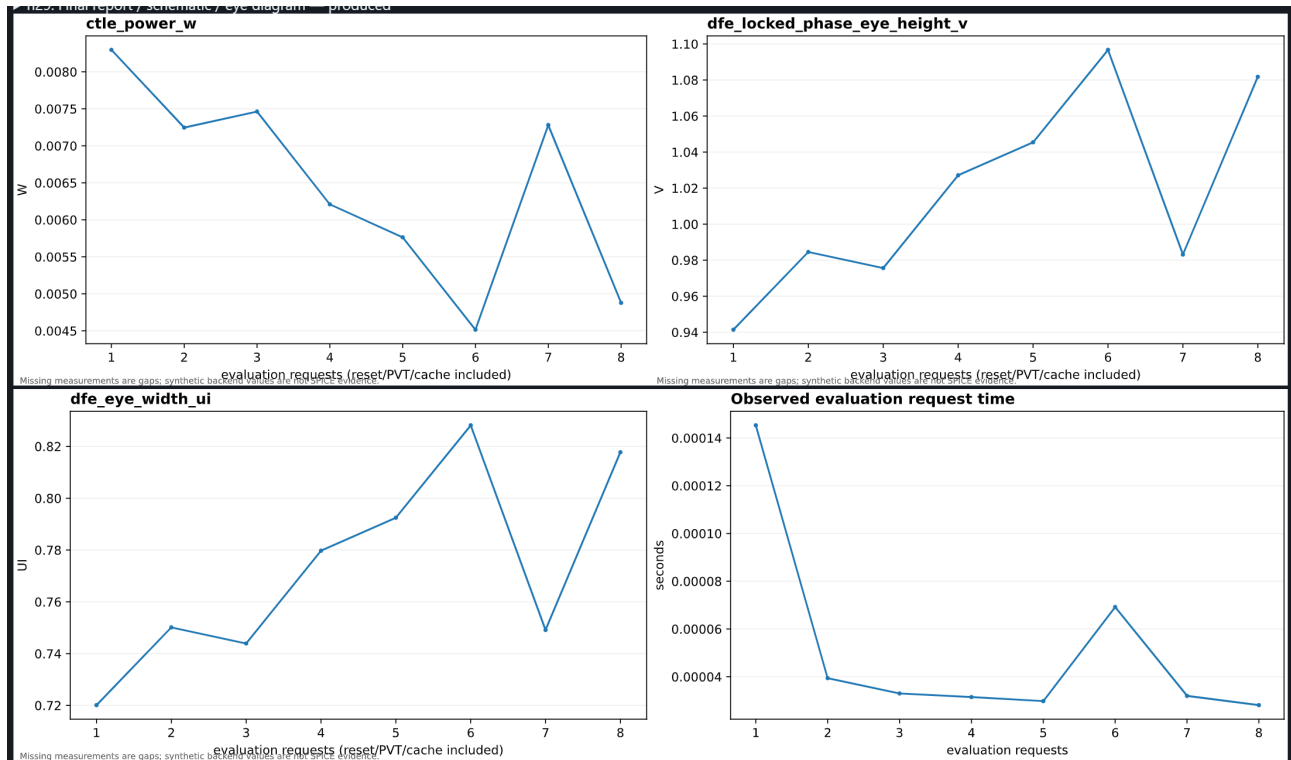
| Metric                 | Value                | Unit  | Condition                        | Source                    | Verification                        |
|------------------------|----------------------|-------|----------------------------------|---------------------------|-------------------------------------|
| HD3                    | -75.62               | dB    | Nominal, 100 mVpp differential   | smoke_evidence_c          | [VERIFIED FROM RAW SIMULATION DATA] |
| Input-referred noise   | 0.372                | mVrms | Nominal, 10 MHz–5 GHz            | Same                      | [VERIFIED FROM RAW SIMULATION DATA] |
| Historical HD3 range   | -79.8428 to -69.6987 | dB    | 36 candidate-fidelity conditions | v3 qualification artifact | [VERIFIED FROM RAW SIMULATION DATA] |
| Historical noise range | 0.328262 to 0.48729  | mVrms | 36 candidate-fidelity conditions | Same                      | [VERIFIED FROM RAW SIMULATION DATA] |

## 19.4 Figure status

The eye PNG/SVG is available. [NOT AVAILABLE IN THE PROVIDED EVIDENCE] A retained raw AC trace suitable for a new report-specific gain curve is not available for every result discussed. Dashboard screenshots of the rendered eye, PVT page, and run graph are not available as browser screenshots in the supplied evidence (see §20.3–§20.4). The supplied PPT/evidence package does contain regenerated analysis charts covering AC response, PVT heatmaps, temperature sensitivity, and metric/runtime evolution, built from stored 27-corner and per-request result data; these are included in the report-local figures/ directory with captions and are labelled as presentation/regenerated-artifact evidence rather than fresh reruns. Several of the supplied filenames (for example the two containing the word "dashboard") do not describe an application screenshot — each image was opened and inspected directly before captioning, and captions below describe what is actually visible, not what the filename implies.



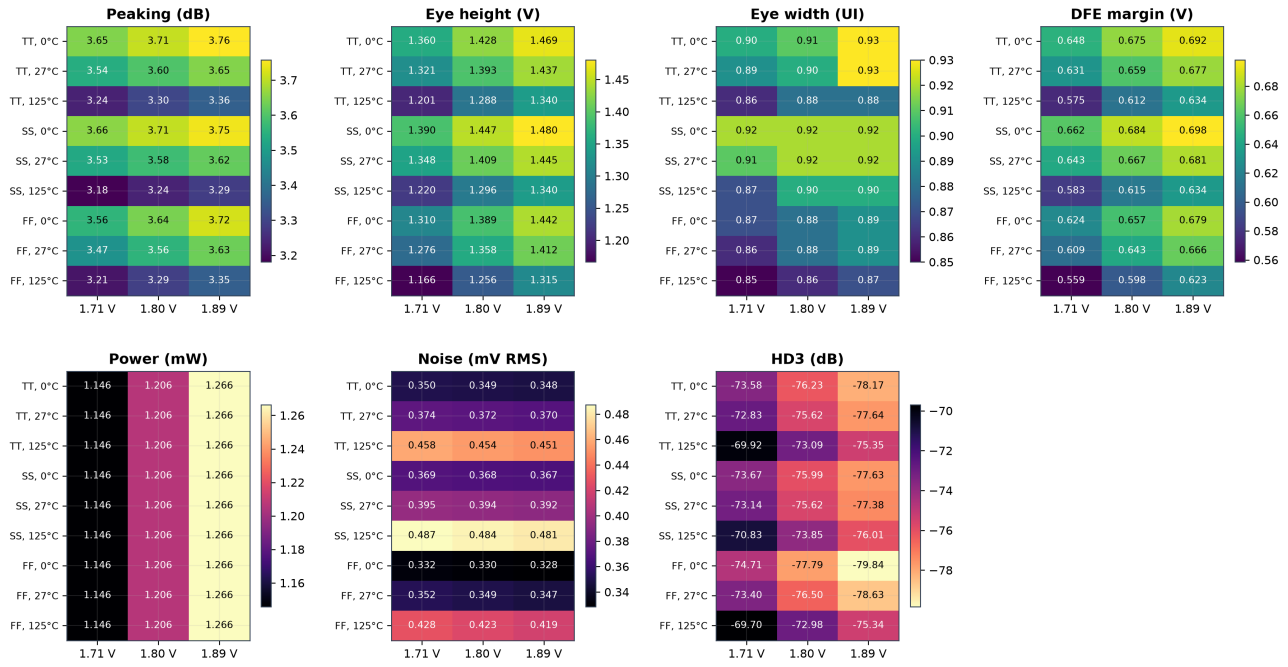
**Figure 5.** Repository-backed receiver eye artifact separating the transistor-level CTLE output eye from the behavioral one-tap DFE sampling eye. **Evidence:** results/v3\_tt\_1000\_eye.png; [VERIFIED FROM RAW SIMULATION DATA]. The DFE panel contains corrected sample points, not a physical DFE waveform.



**Figure 6.** Captured metric and evaluation-time plots (CTLE power, locked- phase eye height, eye width, and observed request time) against successive evaluation requests, including reset/PVT/cache-hit requests. **Evidence:** report\_generation/figures/ppt\_metric\_and\_runtime\_plots.png; [REPORTED IN PPT]. The source image's own caption states that missing measurements and synthetic values are not SPICE evidence; the visible header strip indicates the image was cropped from a larger produced-report/schematic view.

## Measured performance across the completed 512-bit, 27-corner run

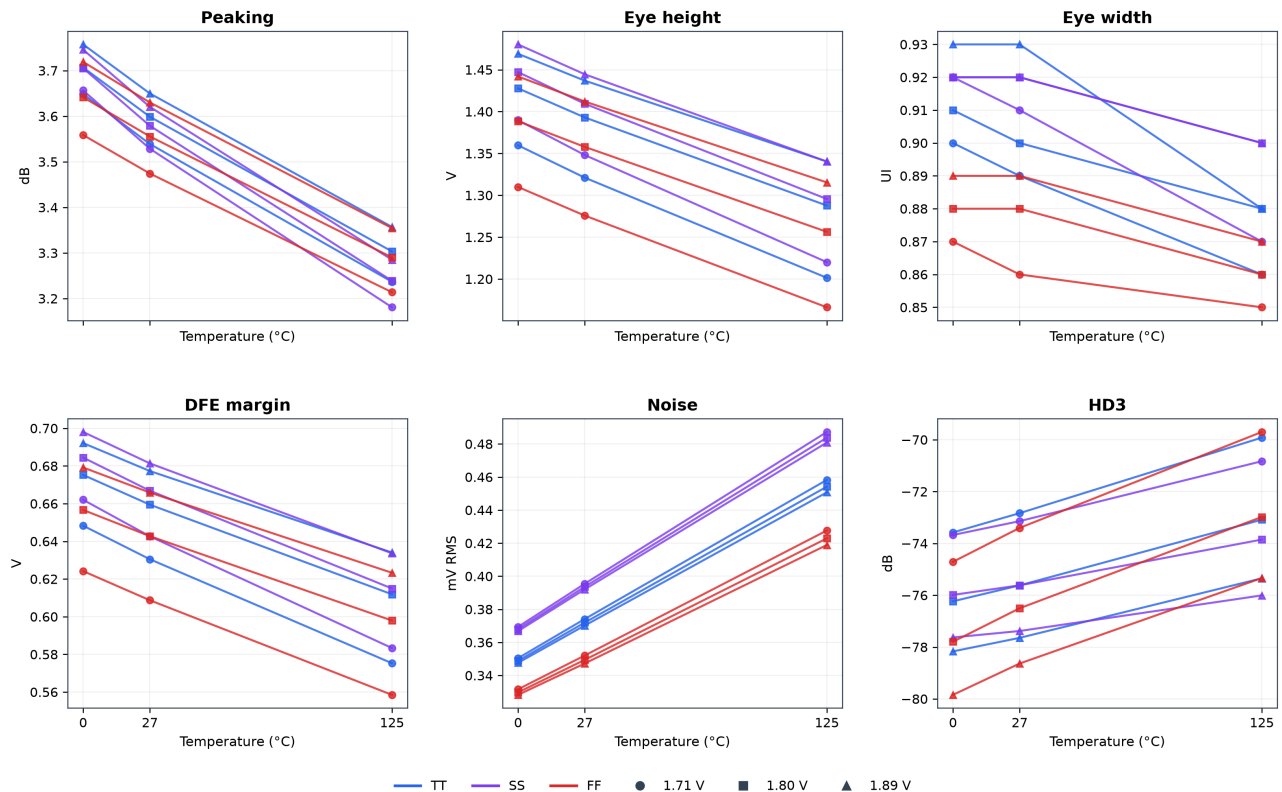
Run 07acbab... • all 27 evaluated conditions passed



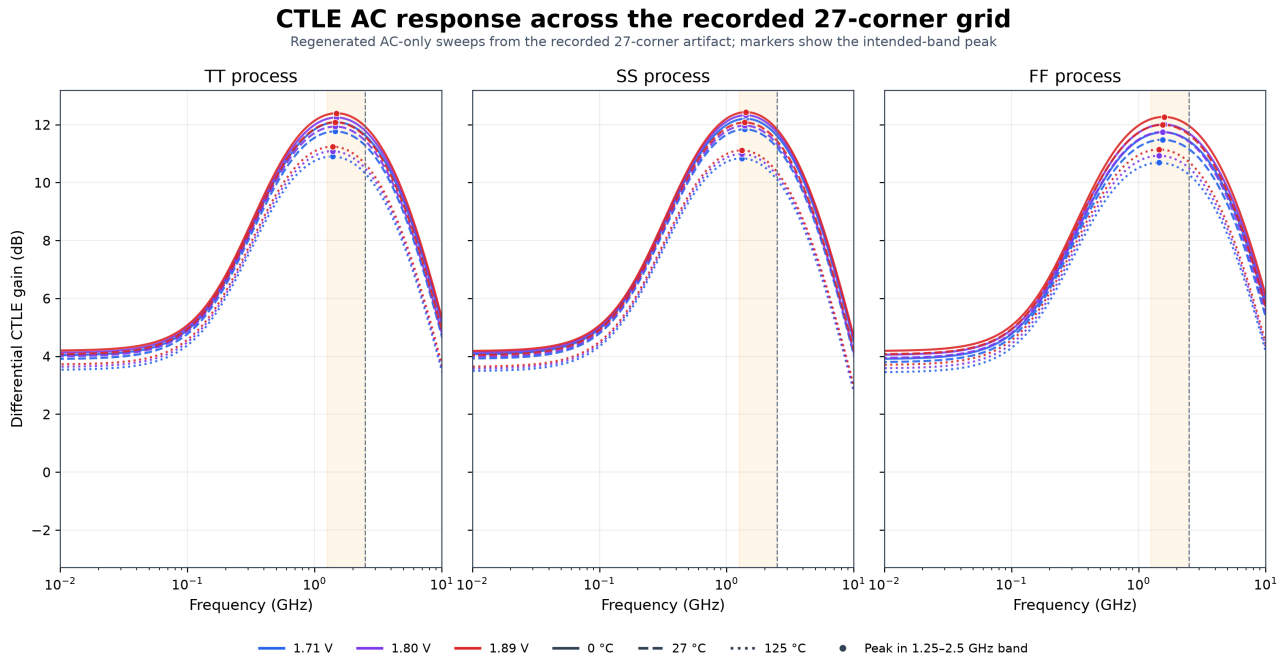
**Figure 7.** Regenerated 27-condition (TT/SS/FF x 1.71/1.80/1.89 V x 0/27/125 degC) performance heatmaps for peaking, eye height, eye width, DFE margin, power, noise, and HD3, titled "Measured performance across the completed 512-bit, 27-corner run" in the source image. **Evidence:** report\_generation/figures/ppt\_dashboard\_overview.png; [REPORTED IN PPT]. Despite its filename, this image is a regenerated analysis chart, not an application/dashboard screenshot — confirmed by direct inspection. It is evidence of the 27-point grid only, not of SF/FS coverage or the full 60-condition grid, and not of a completed FINAL-fidelity rerun. Two further supplied images (ppt\_ac\_corner\_sensitivity.png, ppt\_ac\_peak\_sensitivity.png) regenerate the peaking/peak-frequency slice of this same 27-condition data from separate snapshots and are consistent with the values shown here; they are not reproduced as separate figures to avoid near-duplicate content, but remain available in report\_generation/figures/ as supplementary evidence.

## Temperature sensitivity across process and supply

Colour = process corner; marker = supply voltage



**Figure 8.** Regenerated temperature sensitivity of peaking, eye height, eye width, DFE margin, noise, and HD3 across the TT/SS/FF corners and three supply voltages, at 0/27/125 degC. **Evidence:** report\_generation/figures/ppt\_dashboard\_detail.png; [REPORTED IN PPT]. Despite its filename, this image is also a regenerated analysis chart, not an application/dashboard screenshot. It is drawn from the same 27-condition result set as Figure 7 and does not extend coverage to 75 degC, SF, or FS.



**Figure 9.** AC gain-versus-frequency curves for the TT/SS/FF corners at three supply voltages and three temperatures, regenerated from the recorded 27-corner artifact, with the intended 1.25-2.5 GHz band and peak marked. **Evidence:** report\_generation/figures/ppt\_ac\_response\_27\_corners.png; [REPORTED IN PPT]. This is regenerated historical-artifact evidence, not a fresh rerun; a second, separately regenerated version of the same underlying chart type is also present in figures/ (ppt\_temperature\_sensitivity.png, which — despite its filename — is confirmed by inspection to be an AC response chart, not a temperature-sensitivity chart) and is not reproduced here to avoid duplicating Figure 9.

## 20. Dashboard and LLM Wrapper

### 20.1 Dashboard architecture

[VERIFIED FROM SOURCE CODE] experiments/web\_ui.py exposes a local interface for selecting backend, checkpoint, RL version, channel, target, PVT options, HD3/noise options, and output artifacts. It launches the pipeline as a subprocess instead of importing the core simulator and policy code into the web server.

The UI supports circuit generation and PPO checkpoint training. It exposes run graphs, plots, JSON results, schematics, and eye artifacts when those artifacts were produced by the underlying run.

### 20.2 Natural-language wrapper

[VERIFIED FROM SOURCE CODE] nebula/llm\_wrapper.py parses a user request, constructs the pipeline command, runs it as a subprocess, and returns the structured output. Deterministic parsing is available through the local parser; an optional Anthropic provider is also implemented with fallback behavior.

The wrapper does not run a separate metric model. It does not calculate eye height, invent a PVT verdict, or replace the deterministic final-specification assessment. [ENGINEERING INTERPRETATION] This division keeps natural-language convenience separate from electrical truth.

### 20.3 Interface evidence

| Interface feature                                                                        | Evidence                                               | Status                                   |
|------------------------------------------------------------------------------------------|--------------------------------------------------------|------------------------------------------|
| Generate and verify circuits                                                             | experiments/web_ui.py and wrapper outputs              | [VERIFIED FROM SOURCE CODE]              |
| Train a v3 checkpoint                                                                    | UI command construction and training path              | [VERIFIED FROM DOCUMENTATION]            |
| Download graphs and JSON                                                                 | Results and graph artifacts                            | [VERIFIED FROM RAW SIMULATION DATA]      |
| Final-parameter and measured-specification card view                                     | Figure 4 (§16.5)                                       | [REPORTED IN PPT]                        |
| Browser screenshot of the control panel<br>(target/backend/channel/checkpoint selectors) | No such screenshot found in the supplied evidence      | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| Browser screenshots for every screen<br>(run graph, PVT screen, rendered eye)            | No complete screenshot set                             | [NOT AVAILABLE IN THE PROVIDED EVIDENCE] |
| LLM-generated electrical verdict                                                         | Not used; deterministic pipeline remains authoritative | [VERIFIED FROM SOURCE CODE]              |

## 20.4 What the supplied images establish

Every image supplied for this report was opened and inspected directly before being described here. Two of the supplied filenames contain the word "dashboard" (ppt\_dashboard\_overview.png, ppt\_dashboard\_detail.png), which could be read as implying a captured application screenshot showing the control panel described in experiments/web\_ui.py (target selection, backend selection, channel choice, checkpoint selection). Direct inspection shows neither image is that: both are regenerated matplotlib analysis charts (the 27-condition PVT heatmap grid and the temperature-sensitivity chart reproduced as Figures 7 and 8 in §19.4). No browser screenshot of the control panel itself is present in the supplied evidence. [NOT AVAILABLE IN THE PROVIDED EVIDENCE] This is recorded as missing evidence, not implied by a filename.

[REPORTED IN PPT] The one supplied image that does show genuine application interface chrome is ppt\_final\_parameters\_specification.png (Figure 4, §16.5): labelled parameter fields and a measured-specification table with PASS/NOT CLAIMED verdicts. It establishes that the interface can render a final-parameter card and a verdict table for a captured run. It does not establish that every other displayed chart in this report was generated by that same run, that any browser view was captured during the wrapper screenshot's 412.90 s execution, or that the LLM itself designed or validated a transistor circuit. [VERIFIED FROM SOURCE CODE] The repository remains the primary evidence that the wrapper shells out to experiments.run\_autockt\_pipeline and that the deterministic pipeline computes metrics and verdicts.

## 21. Verification and Test Coverage

### 21.1 Live regression status

[VERIFIED FROM LOGS] The live command `python -m pytest -q tests/` completed with 632 passed, 2 skipped, and 27 subtests passed. Older documents report 419 and 582 tests because the suite grew; those historical counts are not used as the current total.

### 21.2 Test categories

| Area               | Representative tests                          | What is checked                                  |
|--------------------|-----------------------------------------------|--------------------------------------------------|
| Circuit and parser | test_ctle.py, test_parser.py, test_ngspice.py | Parameter handling, parsing, simulator contracts |

| Area                  | Representative tests                                                    | What is checked                                 |
|-----------------------|-------------------------------------------------------------------------|-------------------------------------------------|
| Receiver              | test_receiver_components.py,<br>test_evaluate_pvt_grid.py               | Stage metrics and PVT control flow              |
| PPO v1/v2             | test_autockt_rl.py,<br>test_rl_ppo_v2.py                                | Legacy behavior, validity, reward, truncation   |
| Checkpoints           | test_rl_checkpoint.py,<br>test_v3_integration.py                        | Save/load, compatibility, metadata              |
| Synthetic learning    | test_synthetic_benchmark.py,<br>test_rl_ppo_v2_ablation.py              | Mechanics without ngspice                       |
| Baselines             | test_train_cem.py,<br>test_benchmark_report.py                          | Search mechanics and result consolidation       |
| PVT and qualification | test_pvt_sweep.py,<br>test_qualification_sequence.py                    | Condition sets, stopping, qualification gates   |
| Reporting/UI          | test_final_specification.py,<br>test_v3_dashboard.py,<br>test_web_ui.py | Verdicts, plots, interface command construction |
| Artifact integrity    | test_artifact_integrity.py                                              | SHA-256 package membership and contents         |

### 21.3 What tests do not prove

[ENGINEERING INTERPRETATION] Unit and integration tests prove contracts and control flow, not physical validity over all parameter values. Synthetic PPO tests prove learning mechanics on the controlled synthetic landscape, not analog circuit optimization. The optional SKY130 integration test depends on local model-library configuration and can be skipped.

[NOT AVAILABLE IN THE PROVIDED EVIDENCE] Tests do not substitute for silicon, post-layout extraction, jitter tolerance, mismatch/Monte Carlo, or a complete PCIe compliance test suite.

## 22. Limitations and Missing Evidence

### 22.1 Physical-model limitations

- The DFE is behavioral, not transistor-level. [VERIFIED FROM SOURCE CODE]
- The tail current source is ideal in the Stage-1 CTLE. [VERIFIED FROM SOURCE CODE]
- Passive devices are ideal SPICE R/C elements, not layout-extracted devices. [VERIFIED FROM SOURCE CODE]
- Physical sampler, clocking, DFE power, and DFE area are absent. [VERIFIED FROM DOCUMENTATION]
- Total circuit area and total receiver power are not available. [NOT AVAILABLE IN THE PROVIDED EVIDENCE]

### 22.2 Validation limitations

- The public IEEE/IBM channel is not PCI-SIG compliance evidence. [VERIFIED FROM DOCUMENTATION]
- SF and FS corners exist in the 60-point grid but are not part of the demonstrated 36-point qualification. [VERIFIED FROM SOURCE CODE]
- The final-fidelity 36-condition rerun for the fresh design is incomplete. [VERIFIED FROM LOGS]
- Zero finite-pattern errors are not a BER guarantee. [VERIFIED FROM DOCUMENTATION]
- Continuous tuning, jitter, mismatch, and Monte Carlo signoff are not demonstrated. [NOT AVAILABLE IN THE PROVIDED EVIDENCE]

### 22.3 Optimization limitations

- Fair PPO superiority is not demonstrated. [VERIFIED FROM LOGS]
- Multi-seed real-SPICE optimizer statistics are missing. [NOT AVAILABLE IN THE PROVIDED EVIDENCE]
- Arbitrary-start reliable convergence is not demonstrated. [VERIFIED FROM DOCUMENTATION]
- Surrogate-assisted and multi-actor PPO are not implemented. [VERIFIED FROM DOCUMENTATION]

## 22.4 Evidence-management limitations

- Some historical training commands are not recorded verbatim. [NOT AVAILABLE IN THE PROVIDED EVIDENCE]
- Hardware CPU/GPU details are not supplied for all historical runs. [NOT AVAILABLE IN THE PROVIDED EVIDENCE]
- No browser screenshot of the dashboard control panel (target, backend, channel, or checkpoint selection) is present; two supplied files whose names suggest a dashboard capture are, by direct inspection, regenerated analysis charts rather than application screenshots (§19.4, §20.4). [NOT AVAILABLE IN THE PROVIDED EVIDENCE]
- The Phase 2 artifact count and later checksum-ledger count differ by one file and were not independently reconciled. [REQUIRES FURTHER VERIFICATION]

## 23. Conclusions and Required Next Work

### 23.1 Failed Approaches and Lessons Learned

The repository preserves several negative or incomplete outcomes that are useful engineering evidence rather than defects to hide.

| Observed failure or gap                                               | Lesson                                                                                                          | Required next experiment                                                        |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Fair PPO trial reached 0/20 while Random Search reached 1/20          | PPO's current action reachability, horizon, initialization, and state/reward contract do not prove an advantage | Matched multi-seed, budget-matched real-SPICE comparison                        |
| Many candidates failed at early DC/AC stages                          | Early rejection is necessary, but early failures can provide little learning signal                             | Measure failure-stage distributions and compare v1/v2/v3 reward/state contracts |
| Historical 36/36 result covers TT/SS/FF only                          | Passing the agreed grid is not five-corner qualification                                                        | Complete SF/FS conditions in the defined 60-point grid                          |
| Area is only matched-MOS channel area                                 | Partial geometry cannot support total-area compliance                                                           | Add passive, sampler, DFE, routing, and layout-overhead models                  |
| DFE remains behavioral                                                | Receiver eye evidence does not prove a physical DFE                                                             | Implement and simulate a transistor-level sampler/DFE                           |
| Only one of three tuning targets met the recorded $\pm 5\%$ criterion | Nominal sampled tuning is not continuous tuning compliance                                                      | Run receiver/noise/HD3 checks across the required frequency range               |
| Long runs are expensive and crash logs are preserved                  | Runtime and native-tool stability affect reproducibility                                                        | Record machine/toolchain data and run controlled repeated evaluations           |

[VERIFIED FROM DOCUMENTATION] These observations come from preserved result files, audit documents, and source-level failure handling. [ENGINEERING INTERPRETATION] Each next experiment is a proposed closure step, not a claim that the current repository already performs it.

### 23.2 Conclusions

Nebula demonstrates a functioning, versioned analog-design automation flow: target specification enters a PPO or baseline search, candidates are mapped to a parameterized SKY130 CTLE, ngspice produces staged measurements, a behavioral DFE produces receiver-level metrics, and a deterministic reporting layer records PASS, FAIL, or NOT CLAIMED.

[VERIFIED FROM SOURCE CODE] The project has clear ownership boundaries among optimization, simulation, analysis, and interfaces. [VERIFIED FROM RAW SIMULATION DATA] Real-SPICE nominal and historical qualification artifacts exist. [VERIFIED FROM LOGS] The current software regression suite passes.

The strongest defensible result is therefore an integrated Stage-1 demonstration, not a complete physical receiver product. One historical candidate passed the agreed 36-condition TT/SS/FF grid at candidate fidelity, while another finalist failed part of that grid. Fresh nominal evidence shows successful eye, power, HD3, noise, and finite-pattern error metrics for one selected run. Neither result establishes total chip area, physical DFE operation, silicon performance, or PCIe compliance.

## 23.3 Required next work

The following work is required to strengthen the engineering claim:

1. Implement a physical sampler and transistor-level DFE, including power and area accounting.
2. Replace ideal passive assumptions with documented PDK or extracted layout models.
3. Complete a defensible total-area model and retain the NOT CLAIMED gate until all blocks and layout overhead are represented.
4. Run matched, multi-seed, budget-matched PPO/Random Search/CEM comparisons with wall-clock and raw metric records.
5. Complete the full 60-condition PVT grid, or explicitly redefine the qualification requirement with external approval.
6. Add PRBS15, jitter, mismatch, Monte Carlo, and long-run error testing.
7. Establish continuous tuning evidence across the required frequency range.
8. Obtain approved PCIe channel and compliance-test evidence.
9. Record original training commands, machine information, and complete dashboard screenshots for future reproducibility.

[VERIFIED FROM DOCUMENTATION] These are repository-documented deferred items. They are proposals for completion, not claims that the current repository already implements them.

## References

### Internal repository documents cited in this report

- docs/V3\_QUALIFICATION\_RESULTS\_20260913.md — historical 36-condition PVT qualification results for pipeline\_ep0/pipeline\_ep1 (§16, §17).
- docs/AGREED\_QUALIFICATION.md — agreed qualification scope and simulated-bit-count record (§16, Appendix E).
- docs/FINAL\_TECHNICAL\_AUDIT.md — second qualified/near-qualified design record and SIGSEGV crash-log references (§16, §21).
- docs/RL\_PPO\_V2\_STUDY.md — PPO v2 root-cause and state/reward design notes (§12).
- docs/V3\_TRAINING\_AND\_RUN\_GRAPHS.md — v3 training/run-graph documentation and an older test-suite snapshot (§13, §21, Appendix F).
- docs/PHASE7\_COMPLETION\_REPORT.md — an older test-suite snapshot (§21).
- docs/stage1-decisions.md — deferred Stage-2 scope (§23).
- artifacts/PHASE2\_ARTIFACT\_INDEX.md — Phase 2 artifact package file count, used in the file-count discrepancy note (Appendix F).
- results/ppo\_v3\_tt\_1000\_policy.pt, results/v3\_tt\_1000\_summary.json — genuine v3 checkpoint and training summary (§13, §18).
- results/benchmark\_report.json — fair PPO/Random Search/CEM head-to-head comparison (§14, §18).
- results/baseline\_comparison\_summary.json — non-controlled Random Search/CEM comparison (§14).
- results/v3\_qualification\_final\_20260913.json — historical PVT qualification artifact (§17).
- results/llm\_wrapper\_runs/smoke\_evidence\_demo.json — fresh real-SPICE nominal and smoke-evidence run (§16, §19).

**Primary source modules cited throughout** — see Appendix A for the full index; principal entry points are experiments/run\_autockt\_pipeline.py, simulator/receiver.py::evaluate\_receiver, simulator/config.py, simulator/design\_schema.py, rl/autockt\_v3.py, nebula/llm\_wrapper.py, and analysis/final\_specification.py.

### External standards and tools referenced for context

- SkyWater Foundry / Google open-source SKY130 process design kit — source of the `sky130_fd_pr__nfet_01v8` device model used in `circuits/blocks/ctle.spice` (§6). Cited as the device-model source only; this report makes no claim of PDK-vendor validation or sign-off.
- ngspice, the open-source circuit simulator invoked via `simulator/ngspice.py` (§8–§9, §15). The supplied evidence does not pin a specific ngspice release number.
- Touchstone (SNP) S-parameter file format — used for the four-port channel models parsed by `simulator/channel.py` (§5).
- IEEE 802.3 — source naming convention for the supplied `ieee802_ibm_20db_thru.s4p` reference channel (§5); its own metadata states it is not PCIe compliance evidence, and this report repeats that caveat rather than asserting protocol conformance.
- PCI Express (PCIe) Gen-2 signaling context (5 GT/s, 2.5 GHz Nyquist) — used only to justify the bit-rate default in `simulator/stimulus.py` (§2, §5). This report does not cite a specific PCI-SIG specification revision and makes no claim of electrical, protocol, or compliance-test conformance to it (§22).

## 24. Appendices

### Appendix A. Source and artifact index

| Evidence group           | Primary paths                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| CTLE netlist             | <code>circuits/blocks/ctle.spice</code>                                                                               |
| SPICE benches            | <code>circuits/benches/</code>                                                                                        |
| Receiver evaluator       | <code>simulator/receiver.py</code>                                                                                    |
| Channel parser           | <code>simulator/channel.py</code>                                                                                     |
| Waveform metrics         | <code>simulator/waveform.py</code>                                                                                    |
| DFE and eyes             | <code>simulator/receiver_metrics.py</code>                                                                            |
| Parameter schema         | <code>simulator/design_schema.py</code> ,<br><code>simulator/config.py</code>                                         |
| PPO v1/v2/v3             | <code>rl/autockt_*.py</code> , <code>rl/ppo_agent.py</code> , <code>rl/trainer.py</code>                              |
| Pipeline                 | <code>experiments/run_autockt_pipeline.py</code>                                                                      |
| Baselines                | <code>experiments/receiver_search.py</code> ,<br><code>experiments/train_cem.py</code>                                |
| Analysis/reporting       | <code>analysis/</code>                                                                                                |
| Tests                    | <code>tests/</code>                                                                                                   |
| Fresh results            | <code>results/llm_wrapper_runs/</code> ,<br><code>report_generation/verified_results.md</code>                        |
| Historical qualification | <code>results/v3_qualification_final_20260913.json</code> ,<br><code>docs/V3_QUALIFICATION_RESULTS_20260913.md</code> |
| Report-local figures     | <code>report_generation/figures/</code>                                                                               |

### Appendix B. Evidence-status vocabulary

| Label                                    | Meaning                                         |
|------------------------------------------|-------------------------------------------------|
| [VERIFIED FROM SOURCE CODE]              | Direct implementation or test evidence          |
| [VERIFIED FROM RAW SIMULATION DATA]      | Value read from a result artifact               |
| [VERIFIED FROM LOGS]                     | Value confirmed by command or run logs          |
| [REPORTED IN PPT]                        | Presentation context not sufficient by itself   |
| [VERIFIED FROM DOCUMENTATION]            | Explicit project documentation evidence         |
| [ENGINEERING INTERPRETATION]             | Technical interpretation, not a raw measurement |
| [NOT AVAILABLE IN THE PROVIDED EVIDENCE] | No supplied supporting artifact                 |
| [REQUIRES FURTHER VERIFICATION]          | Partial, historical, or unresolved evidence     |

## Appendix C. Important numerical results

| Metric           | Value                 | Unit       | Source                    | Verification Status                 | Interpretation              |
|------------------|-----------------------|------------|---------------------------|-------------------------------------|-----------------------------|
| Fresh eye height | 1.55                  | V          | smoke_evidence_c          | [VERIFIED FROM RAW SIMULATION DATA] | Nominal selected run        |
| Fresh eye width  | 0.73                  | UI         | Same                      | [VERIFIED FROM RAW SIMULATION DATA] | Nominal selected run        |
| Fresh CTLE power | 1.206                 | mW         | Same                      | [VERIFIED FROM RAW SIMULATION DATA] | CTLE-level only             |
| Fresh HD3        | -75.62                | dB         | Same                      | [VERIFIED FROM RAW SIMULATION DATA] | 100 mVpp nominal test       |
| Fresh noise      | 0.372                 | mVrms      | Same                      | [VERIFIED FROM RAW SIMULATION DATA] | 10 MHz–5 GHz integration    |
| Historical PVT   | 36/36                 | conditions | v3 qualification artifact | [VERIFIED FROM RAW SIMULATION DATA] | TT/SS/FF candidate fidelity |
| Fair PPO         | 0/20                  | successes  | benchmark report          | [VERIFIED FROM LOGS]                | One-seed comparison         |
| Live tests       | 632 passed, 2 skipped | tests      | pytest output             | [VERIFIED FROM LOGS]                | Current regression result   |

## Appendix D. Figure register

Figures are numbered in document order. "Content verified by direct inspection" means the image file was opened and viewed during this reporting pass and its caption describes what is actually visible, independent of its filename.

| Figure | Section | Content                                                                                               | Source file                               | Status                                                                                                                                                |
|--------|---------|-------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | 6       | CTLE schematic (matched MOS pair, loads, RC degeneration, tail source)                                | figures/ppt_ctle_to                       | Supplied PPT asset; content verified by direct inspection                                                                                             |
| 2      | 9       | Optimization and verification flow (candidate search; selection and verification)                     | figures/ppt_optimiz                       | Supplied PPT asset; content verified by direct inspection                                                                                             |
| 3      | 16      | Working-wrapper console result (text transcript only; no image file)                                  | User-supplied attachment "Pasted text #1" | Image not mounted as a local project asset; see unavailable-figure notice                                                                             |
| 4      | 16      | Final circuit-parameter cards and measured-specification table (PASS/NOT CLAIMED)                     | figures/ppt_final_p                       | Supplied PPT asset; the one image confirmed to show genuine application interface chrome                                                              |
| 5      | 19      | Receiver eye diagram (CTLE output eye and behavioral DFE sampling eye)                                | results/v3_tt_1000_                       | Repository-backed raw artifact                                                                                                                        |
| 6      | 19      | Metric and runtime plots vs. evaluation request index                                                 | figures/ppt_metric_                       | Supplied PPT asset; content verified by direct inspection                                                                                             |
| 7      | 19      | 27-condition PVT performance heatmaps (peaking, eye height, eye width, DFE margin, power, noise, HD3) | figures/ppt_dashboa                       | Supplied asset; filename says "dashboard" but content verified by direct inspection to be a regenerated analysis chart, not an application screenshot |
| 8      | 19      | Temperature sensitivity (peaking, eye height, eye width, DFE margin, noise, HD3 vs. temperature)      | figures/ppt_dashboa                       | Supplied asset; filename says "dashboard" but content verified by direct inspection to be a regenerated analysis chart, not an application screenshot |

| Figure | Section | Content                                                    | Source file         | Status                                                    |
|--------|---------|------------------------------------------------------------|---------------------|-----------------------------------------------------------|
| 9      | 19      | AC gain vs. frequency across the 27-corner grid (TT/SS/FF) | figures/ppt_ac_resp | Supplied PPT asset; content verified by direct inspection |

Supplementary, non-figure-numbered images present in `report_generation/figures/` but not separately reproduced as report figures (to avoid near-duplicate content already covered by Figures 7 and 9): `ppt_ac_corner_sensitivity.png` and `ppt_ac_peak_sensitivity.png` (peaking/peak-frequency slices of the same 27-condition data as Figure 7, from separate regeneration passes) and `ppt_temperature_sensitivity.png` (a second, separately regenerated AC response chart of the same type as Figure 9, despite its filename). All three were opened and inspected directly; none contradicts the figures above.

## Appendix E. Reproduction checklist

1. Use the repository revision and compatible Python environment.
2. Configure `NGSPICE_EXECUTABLE` and `SKY130_MODEL_LIBRARY`.
3. Select the correct RL version and metadata-compatible checkpoint.
4. Record target values, channel path, port map, fidelity, PVT condition set, seed, horizon, and evaluation budget.
5. Run the requested pipeline and retain JSON/JSONL outputs and provenance.
6. Treat synthetic backend results as software-mechanics evidence only.
7. Do not promote partial area, incomplete PVT, or finite-pattern zero errors into stronger claims.
8. Compare any new result with the source-linked evidence tables before updating a conclusion.

## Appendix F. Unavailable information register

The following remain explicitly unavailable in the provided evidence:

- complete receiver/chip area;
- total receiver power;
- transistor-level sampler and DFE results;
- post-layout extraction and DRC closure;
- silicon or commercial-SPICE cross-validation;
- full five-corner 60-condition completed qualification;
- multi-seed statistically powered optimizer superiority;
- guaranteed BER, jitter, mismatch, and Monte Carlo signoff;
- continuous tuning compliance;
- complete dashboard screenshot set;
- original v3 training command and all machine specifications.

These omissions are limitations of the evidence base, not values to be estimated from unrelated artifacts.

## Appendix G. Evidence and Reproducibility Notes

The report distinguishes source-code contracts, raw result artifacts, command logs, historical documentation, and presentation screenshots. Source code defines implemented behavior; raw JSON/JSONL and waveform artifacts support measured values; logs support execution and timing; documentation records historical scope and limitations; screenshots support only what is visibly shown. These evidence classes are not interchangeable.

Reproducible claims should identify the repository revision, netlists and model library, channel and port map, checkpoint/schema, seed, fidelity, operating condition, target, result artifact, and whether the evidence is fresh, historical, or synthetic. `simulator/provenance.py`, the `ngspice` wrapper, and v3 checkpoint metadata implement much of this identity tracking. Missing fields remain unavailable rather than being reconstructed from screenshots.

[Figure unavailable: the complete combined CTLE schematic-and-parameter image supplied in the final review message is not mounted as a local project asset. The report includes the available CTLE schematic figure and the source-backed parameter table separately in Section 6.]

The 27-point and 36-point PVT artifacts are intentionally reported as different grids: 27 points use TT/SS/FF with 0/27/125 °C, while 36 points use TT/SS/FF with 0/27/75/125 °C. The defined 60-point grid adds SF and FS. This distinction is part of the evidence record, not a presentation preference.